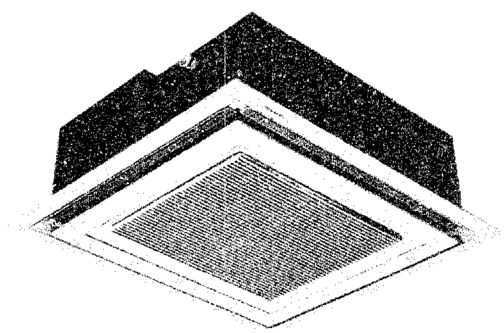
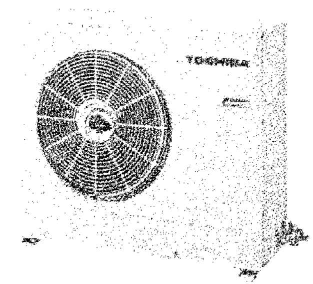
SERVICE DATA
FILE NO. 300-761

TOSHIBA

AIR-CONDITIONER SPLIT (CASSETTE TYPE) HEAT PUMP

RAV-453UHE/S453HE
RAV-713UHE8/S713HE8





Specifications are subject to change without notice.

CONTENTS

1. SPECIFICATIONS

ITEM		MODEL	RAV-453UHE		RAV-713UHE8	
Cooling capacity *1		kcal/h	4,000		6,300	
		BTU/h	16,000		25,200	
		kW	4.7		7.3	
Heating capacity *2		kcal/h	4,300		6,800	
		BTU/h	17,200		27,200	
		kW	5.0		7.9	
Power sourse		Phase	1		3	
		V	220/240		380/415	
		Hz	50		50	
			COOLING	HEATING	COOLING	HEATING
Power consumption		kw	2.0	1.9	2.6	2.5
Power factor		%	90	90	85	86
Running current		A	9.7	9.2	4.4	4.2
Starting current		A	60		28	
Operating noise (SPL)	Indoor unit	dB(A)	50/47/44		50/47/44	
	Outdoor unit	dB(A)	50		53	
Refrigerant	Name of refrigerant		R-22		R-22	
	Charge volume	kg	1.35		2.3	
	Add. volume (5-25m)	g/m	20		35	
Refrigerant control			Capillary tube & Expansion valve		Capillary tube & Expansion valve	
Interconnection pipe	Gas side size	mm (in.)	ϕ 12.7		ϕ 15.9	
	Coupler style		Flare		Flare	
	Liquid side size	mm (in.)	ϕ 6.4		ϕ 9.5	
	Coupler style		Flare		Flare	
	Standard length	m (ft)	5 (16.4′)		5 (16.4′)	
	Maximum length *3 (of one way)	m (ft)	25 (82′)		25 (82′)	
	Maximum height					
	Indoor unit higher	m (ft)	15 (49′)		15 (49′)	
	Outdoor unit higher	m (ft)	15 (49′)		15 (49′)	
Condensate drain pipe diameter		mm	ϕ 32 (OD)		ϕ 32 (OD)	
INDOOR UNIT Model			**RAV-453UHE**		**RAV-713UHE8**	
Appearance colour			Black (Zinc galvanized steel + Thermal insulator)		Black (Zinc galvanized steel + Thermal insulator)	
Dimensions	Height	mm (ft-in.)	300 (11-13/16″)		300 (11-13/16″)	
	Width	mm (ft-in.)	820 (2′8-1/4″)		820 (2′8-1/4″)	
	Depth	mm (ft-in.)	720 (2′4-5/16″)		720 (2′4-5/16″)	
Net weight		kg (lbs)	30 (66)		30 (66)	
Heat exchanger type			Finned tube		Finned tube	
Indoor fan type			Propeller fan		Propeller fan	
Air volume		m^3/h (CFM)	960 (565)		1,260 (742)	
Fan motor output		W	37		37	
CEILING PANEL Model			**RBC-U712PAE**		**RBC-U712PAE**	
Appearance colour			Silky white (Munsell 5Y8.9/1)		Silky white (Munsell 5Y8.9/1)	
Dimensions	Height	mm (ft-in.)	30 (1-3/16″)		30 (1-3/16″)	
	Width	mm (ft-in.)	950 (3′1-3/8″)		950 (3′1-3/8″)	
	Depth	mm (ft-in.)	850 (2′9-7/16″)		850 (2′9-7/16″)	
Net weight		kg (lbs)	8.5 (19)		8.5 (19)	
Air filter			Washable		Washable	
OUTDOOR UNIT Model			**RAV-S453HE**		**RAV-S713HE8**	
Appearance colour			Bronze white (Munsell 6Y7.5/1)		Bronze white (Munsell 6Y7.5/1)	
Dimensions	Height	mm (ft-in.)	740 (2′5-1/8″)		790 (2′7-3/8″)	
	Width	mm (ft-in.)	880 (2′10-5/8″)		880 (2′10-5/8″)	
	Depth	mm (ft-in.)	310 (1′3/16″)		310 (1′3/16″)	
Net weight		kg (lbs)	60 (132)		80 (176)	
Heat exchanger type			Finned tube		Finned tube	
Outdoor fan type			Propeller fan		Propeller fan	
Fan motor output		W	39		63	
Compressor	Model		PH230X3-4LS		YH277JA	
	Output	kW	1.5		2.0	
Safety device			High pressure switch, Fuse Overcurrent relay, Crankcase heater Over load relay, Float switch		High pressure switch, Fuse Overcurrent relay, Crankcase heater Inner over load relay, Float switch	

Specificatons are subject to change without notice.

Note 1: Cooling capacity is based on the follwing temperature conditions.
- Indoor air inlet temperature: 27°C DB (80°F DB) 19.5°C WB (67°F WB)
- Outdoor air inlet temperature: 35°C DB (95°F DB)

Note 2: Heating capacity is based on the follwing temperature conditons.
- Indoor air inlet temperature: 21°C DB (70°F DB)
- Outdoor air inlet temperature: 7°C DB (45°F DB) 6°C WB (43°F WB)

Note 3: These mean actual length.

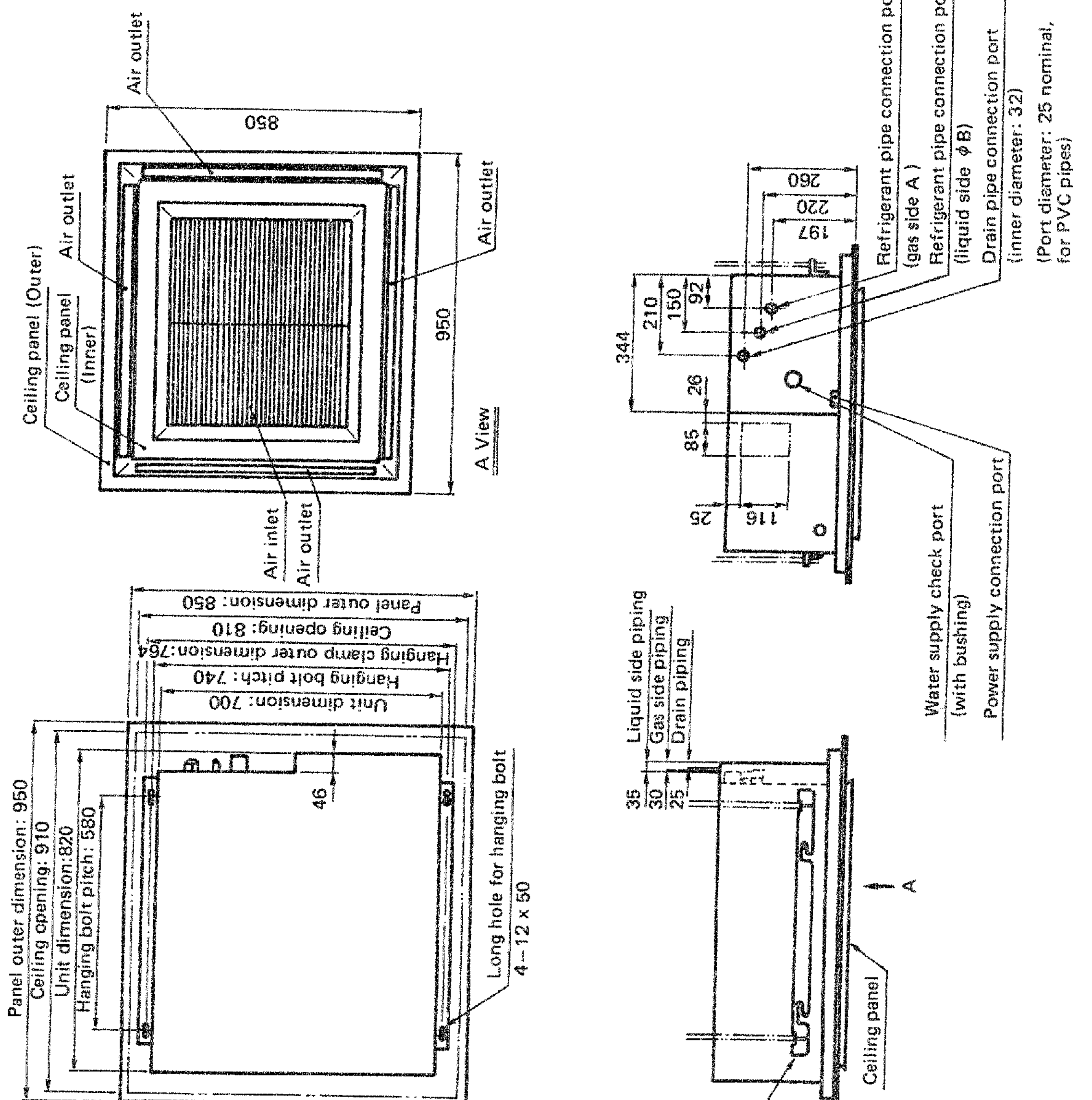
Note 4: Operating range of the units
- Indoor air temperature
  - When cooling
    - Maximum 32°C DB, 22.5°C WB (90°F DB, 73°F WB)
    - Minimum 18°C DB, 15.5°C WB (65°F DB, 60°F WB)
  - When heating
    - Maximum 30°C DB (86°F DB)
    - Minimum 18°C DB (65°F DB)
- Outdoor air temperature
  - When cooling
    - Maximum 43°C DB, 25.5°C WB (109°F DB, 78°F WB)
    - Minimum 10°C DB (50°F DB)
  - When heating
    - Maximum 21°C DB, 15.5°C WB (70°F DB, 60°F WB)
    - Minimum -10°C WB (14°F WB)

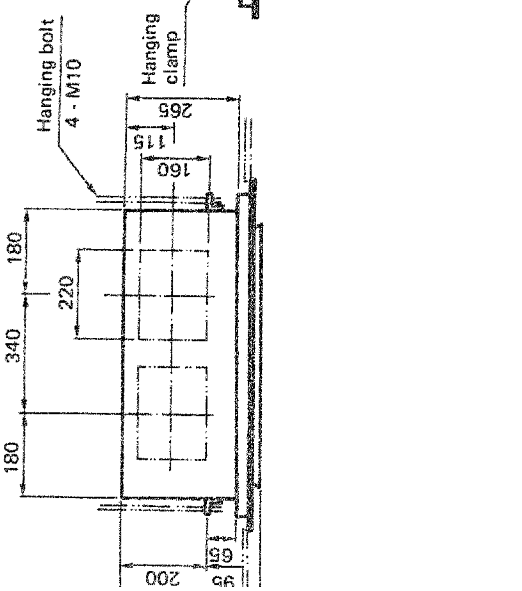
2. CONSTRUCTION VIEWS

2.1 Indoor unit

Unit model
RAV-453UHE
RAV-713UHE8

Panel model
RBC-U712PAE



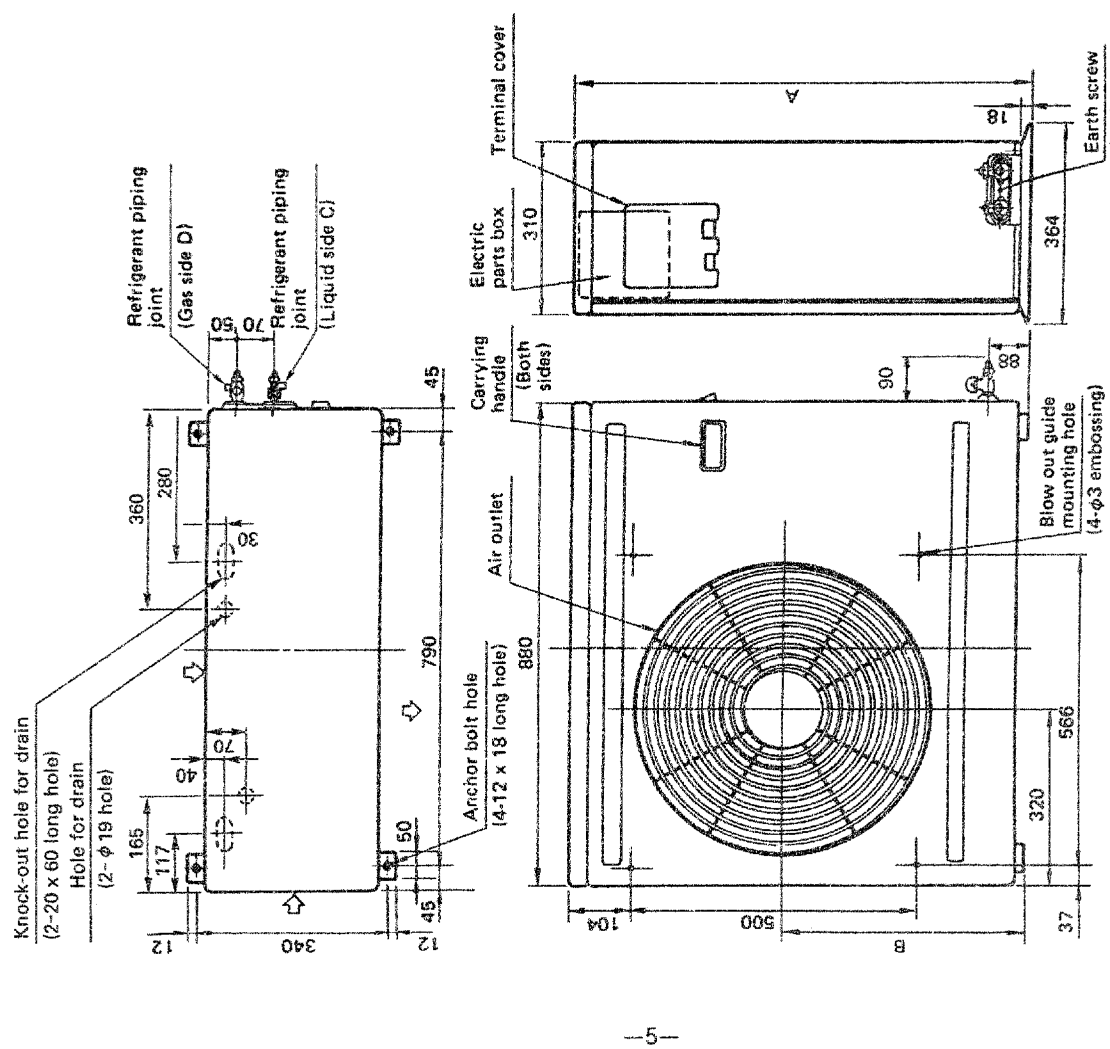


(Unit: mm)

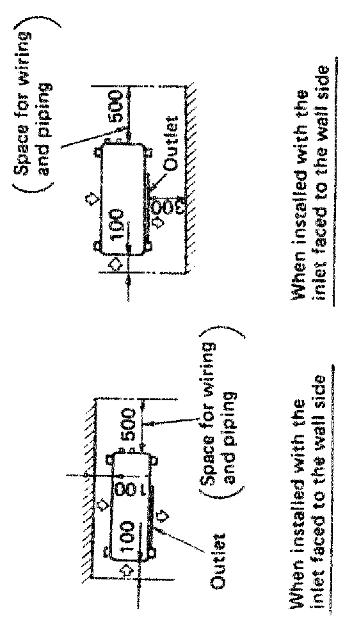
Model	A	B
RAV-453UHE	ϕ12.7	ϕ6.4
RAV-713UHE8	ϕ15.9	ϕ9.5

2.2 Outdoor unit

RAV-S453HE
RAV-S713HE8



Space required for service

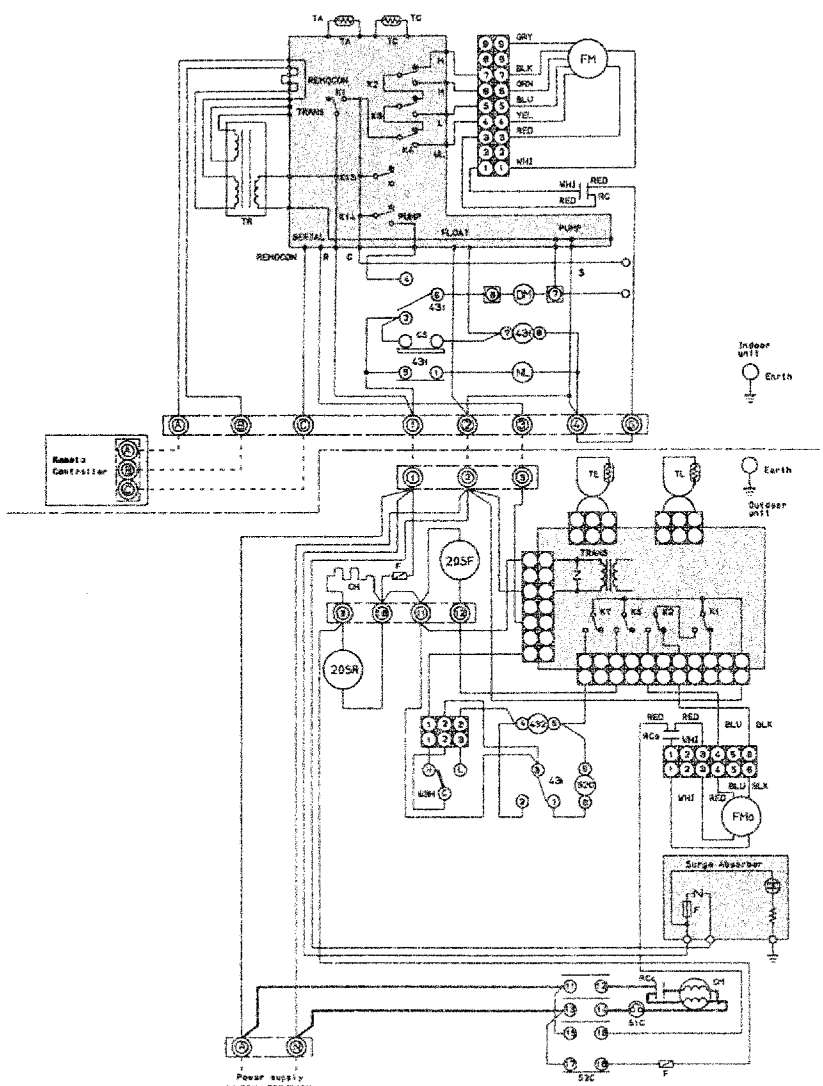


(Unit: mm)

Model	A	B	C	D
RAV-S453HE	740	380	φ6.4	φ12.7
RAV-S713HE8	790	430	φ9.5	φ15.9

3. WIRING DIAGRAM

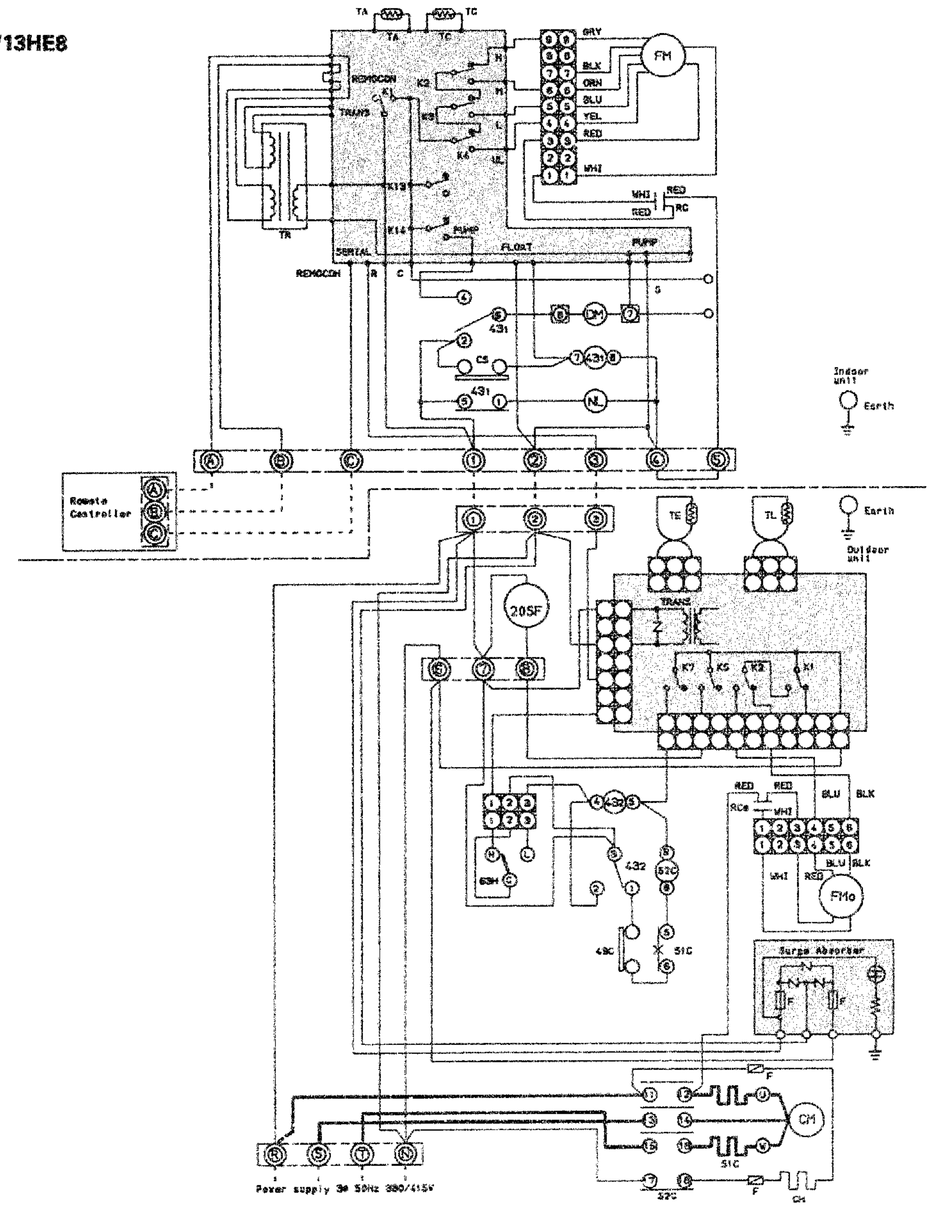
RAV-453UHE/S453HE



◎ shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	**Name**	**Symbol**	**Name**
20SF	Solenoid Coil (4way valve)	CM	Compressor
$K_1 \sim K_{14}$	Relay	52C	Magnetic Contactor
20SR	Solenoid Coil (2way valve)	FM_O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
$43_{1,2}$	Relay	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC_O	Running Capacitor (Outdoor Fan Motor)
FM	Fan Motor (Indoor)		
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor		
TR	Transformer	RC_C	Running Capacitor (Compressor)
CS	Float Switch		
NL	Neon Lamp	DM	Drain Motor

RAV-713UHE8/S713HE8



◎ shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	**Name**	**Symbol**	**Name**
20SF	Solenoid Coil	CM	Compressor
$K_1 \sim K_{14}$	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM_O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
$43_{1,2}$	Relay	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC_O	Running Capacitor (Outdoor Fan Motor)
TA	Sensor		
TC	Sensor	RC	Running Capacitor (Indoor Fan Motor)
FM	Fan Motor (Indoor)		
TR	Transformer	NL	Neon Lamp
CS	Float Switch	DM	Drain Motor

4.SPECIFICATIONS OF ELECTRICAL PARTS

4.1 RAV-453UHE/S453HE

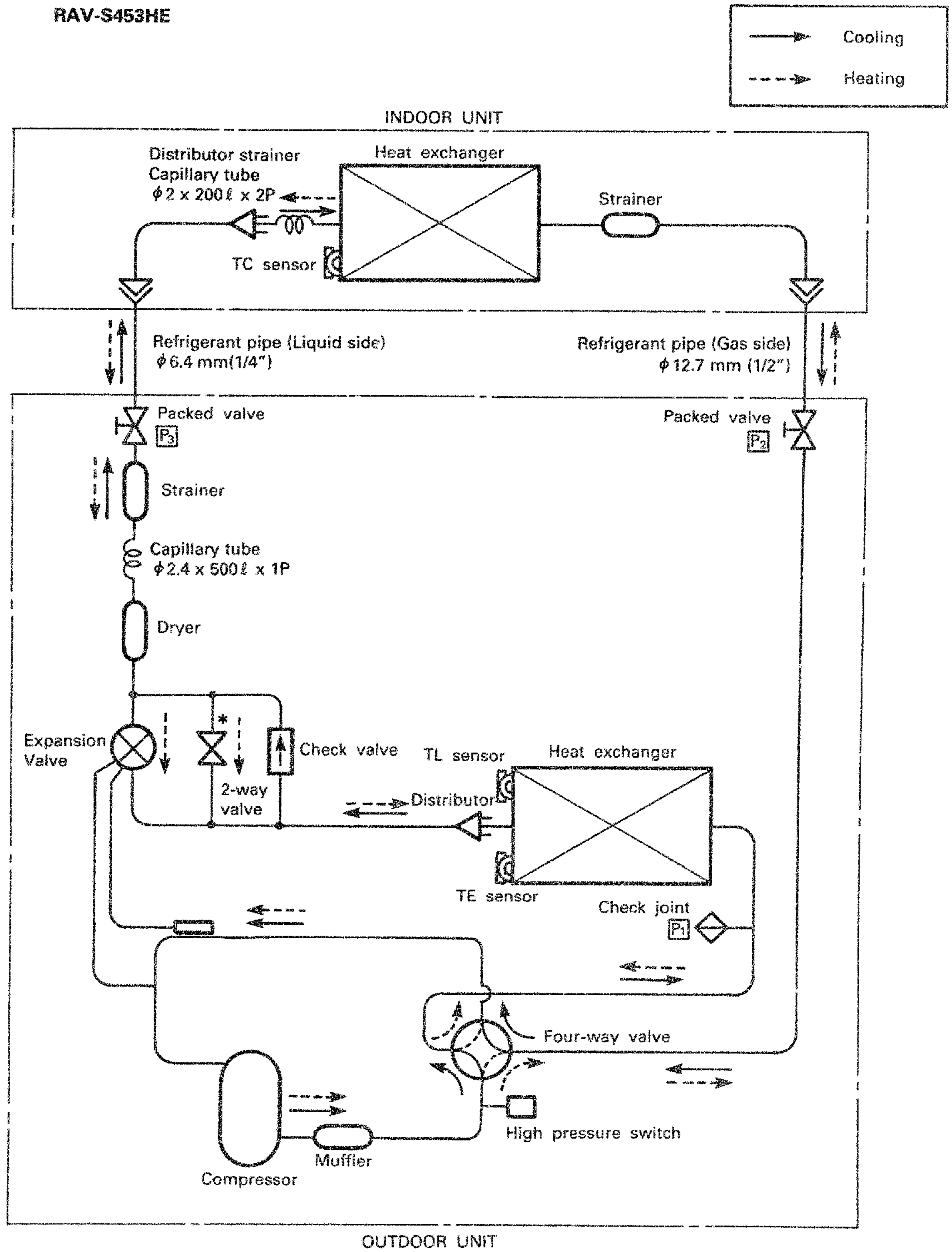
NO.	PARTS NAME	TYPE	SPECIFICATIONS						
1	Indoor unit fan motor	STF-200-44-4H	Output (Rated) 44W, 4pole, 1phase, 200V, 50Hz						
2	Running capacitor for indoor fan motor	40FAFN205UJ	AC400V, 2μF						
3	Transformer	FT12	187～264V						
4	Relay	LY2F-L	AC240V, 2ab						
5	Neon lamp	289-GK-UF	AC240V, Green						
6	Drain pump	PAD-416A23TF	AC230V, 0.2A						
7	Float switch	FS-085-003B	AC230V, 0.5A						
8	Sensor for room temperature		Maximum input 450mW	℃	15	20	25	30	40
				kΩ	16.1	12.6	10.0	8.0	5.2
9	Indoor unit sensor for heat-exchanger temp.	DTN-C103J40	Maximum input 34mA	℃	−12	0	25	50	
				kΩ	62.29	32.82	10.0	3.59	
10	Compressor	PH230X3-4LS	Output (Rated) 1.5kW, 2pole, 1phase, 220/240V, 50Hz						
11	Running capacitor for compressor	MT-42MP456W	AC420V,45μF						
12	Outdoor unit fan motor	AF-230-39P	Output (Rated) 39W, 6pole, 1phase, 230V, 50Hz						
13	Running capacitor for outdoor fan motor	35FAFN405UJ	AC350V, 4μF						
14	Magnetic contactor	E-20F-F	AC230V, 50Hz						
15	High pressure switch	HTB-T317	Tripping pressure 30kg/cm^{2}G Resetting pressure 23kg/cm^{2}G						
16	Solenoid coil for four way valve	L270021	AC240V						
17	Solenoid coil for two way valve		AC220/240V, 50Hz						
18	Relay	SA106H	AC230V 50/60Hz						
19	Crankcase heater		AC240V, 28W						
20	Sensor for defrosting		Maximum input 15.5mA	℃	−12	10			
				kΩ	67.5	21.3			
21	Sensor for coding operation in low ambient temperature	DTN-C103J40	Maximum input 34mA	℃	−12	0	25	50	
				kΩ	62.29	32.82	10.0	3.59	
22	Overload relay	OL-177GM15	AC240V,Tripping temp. 165℃, Resetting temp. 80℃						
23	Fuse		1A						

4.2 RAV-713UHE8/S713HE8

<table>
<tr><th>NO.</th><th>PARTS NAME</th><th>TYPE</th><th colspan="7">SPECIFICATIONS</th></tr>
<tr><td>1</td><td>Indoor unit fan motor</td><td>AF-200-37L</td><td colspan="7">Output (Rated) 37W, 4pole, 1phase, 200V, 50/60Hz</td></tr>
<tr><td>2</td><td>Running capacitor for indoor fan motor</td><td>EEP2G355HQA114</td><td colspan="7">AC400V, 3.5μF</td></tr>
<tr><td>3</td><td>Transformer</td><td>FT12</td><td colspan="7">187~264V</td></tr>
<tr><td>4</td><td>Relay</td><td>LY2F-L</td><td colspan="7">AC240V, 2ab</td></tr>
<tr><td>5</td><td>Neon lamp</td><td>289-GK-UF</td><td colspan="7">AC240V, Green</td></tr>
<tr><td>6</td><td>Drain pump</td><td>PAD-416A23TF</td><td colspan="7">AC230V, 0.2A</td></tr>
<tr><td>7</td><td>Float switch</td><td>FS-085-003B</td><td colspan="7">AC230V, 0.5A</td></tr>
<tr><td rowspan="2">8</td><td rowspan="2">Sensor for room temperature</td><td rowspan="2"></td><td>Maximum input</td><td>℃</td><td>15</td><td>20</td><td>25</td><td>30</td><td>40</td></tr>
<tr><td>450mW</td><td>kΩ</td><td>16.1</td><td>12.6</td><td>10.0</td><td>8.0</td><td>5.2</td></tr>
<tr><td rowspan="2">9</td><td rowspan="2">Indoor unit sensor for heat-exchanger temp.</td><td rowspan="2">DTN-C103J40</td><td colspan="2">Maximum input</td><td>℃</td><td>−12</td><td>0</td><td>25</td><td>50</td></tr>
<tr><td colspan="2">34mA</td><td>kΩ</td><td>62.29</td><td>32.82</td><td>10.0</td><td>3.59</td></tr>
<tr><td>10</td><td>Compressor</td><td>YH277JA</td><td colspan="7">Output (Rated) 3.0kW, 2pole, 3phase, 380/415V, 50Hz</td></tr>
<tr><td>11</td><td>Outdoor unit fan motor</td><td>AF-230-63P</td><td colspan="7">Output (Rated) 63W, 6pole, 1phase, 230V, 50Hz</td></tr>
<tr><td>12</td><td>Running capacitor for outdoor fan motor</td><td>EEP2G405HQA114</td><td colspan="7">AC400V, 4μF</td></tr>
<tr><td>13</td><td>Magnetic contactor</td><td>ME-20F-FS</td><td colspan="7">AC230V, 50Hz</td></tr>
<tr><td>14</td><td>High pressure switch</td><td>HTB-T317</td><td colspan="7">Tripping pressure 30kg/cm²G
Resetting pressure 23kg/cm²G</td></tr>
<tr><td>15</td><td>Solenoid coil</td><td>L270021</td><td colspan="7">AC240V, 50/60Hz</td></tr>
<tr><td>16</td><td>Crankcase heater</td><td></td><td colspan="7">AC240V, 40W</td></tr>
<tr><td rowspan="2">17</td><td rowspan="2">Sensor for defrosting</td><td rowspan="2"></td><td colspan="2">Maximum input</td><td>℃</td><td colspan="2">−12</td><td colspan="2">10</td></tr>
<tr><td colspan="2">15.5mA</td><td>kΩ</td><td colspan="2">67.5</td><td colspan="2">21.3</td></tr>
<tr><td>18</td><td>Overcurrent relay</td><td>RC4-20RSI</td><td colspan="7">Tripping current 7.5A, Resetting auto.</td></tr>
<tr><td>19</td><td>Fuse</td><td></td><td colspan="7">1A</td></tr>
<tr><td>20</td><td>Relay</td><td>SA106H</td><td colspan="7">AC230V, 50/60Hz</td></tr>
<tr><td rowspan="2">21</td><td rowspan="2">Sensor for cooling operation in low ambient temperature</td><td rowspan="2"></td><td colspan="2">Maximum input</td><td>℃</td><td>−12</td><td>0</td><td>25</td><td>50</td></tr>
<tr><td colspan="2">34mA</td><td>kΩ</td><td>62.29</td><td>32.82</td><td>10.0</td><td>3.59</td></tr>
<tr><td>22</td><td>Inner overload relay</td><td>CS-7</td><td colspan="7">Tripping temperature 101℃
Resetting temperature 82℃</td></tr>
</table>

5. REFRIGERANT PIPING DIAGRAM

Indoor unit **Outdoor unit**
RAV-453UHE **RAV-S453HE**



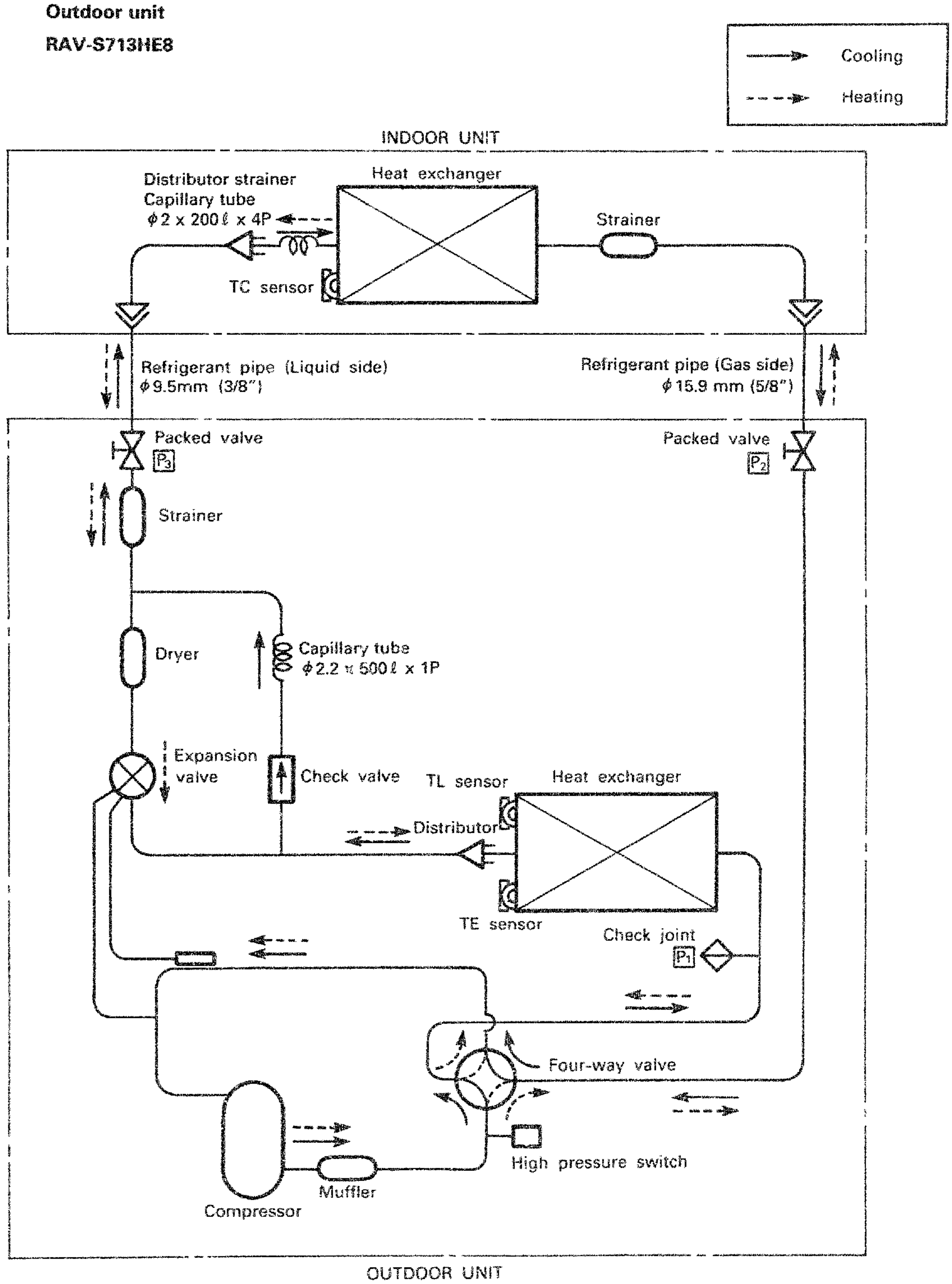
* 2-way valve will open when compressor will go off on Heating operation.

Line Pressure

	Cooling	Heating
P_1	High pressure	Low pressure
P_2	Low pressure	High pressure
P_3	Medium pressure	Medium pressure

Indoor unit **Outdoor unit**

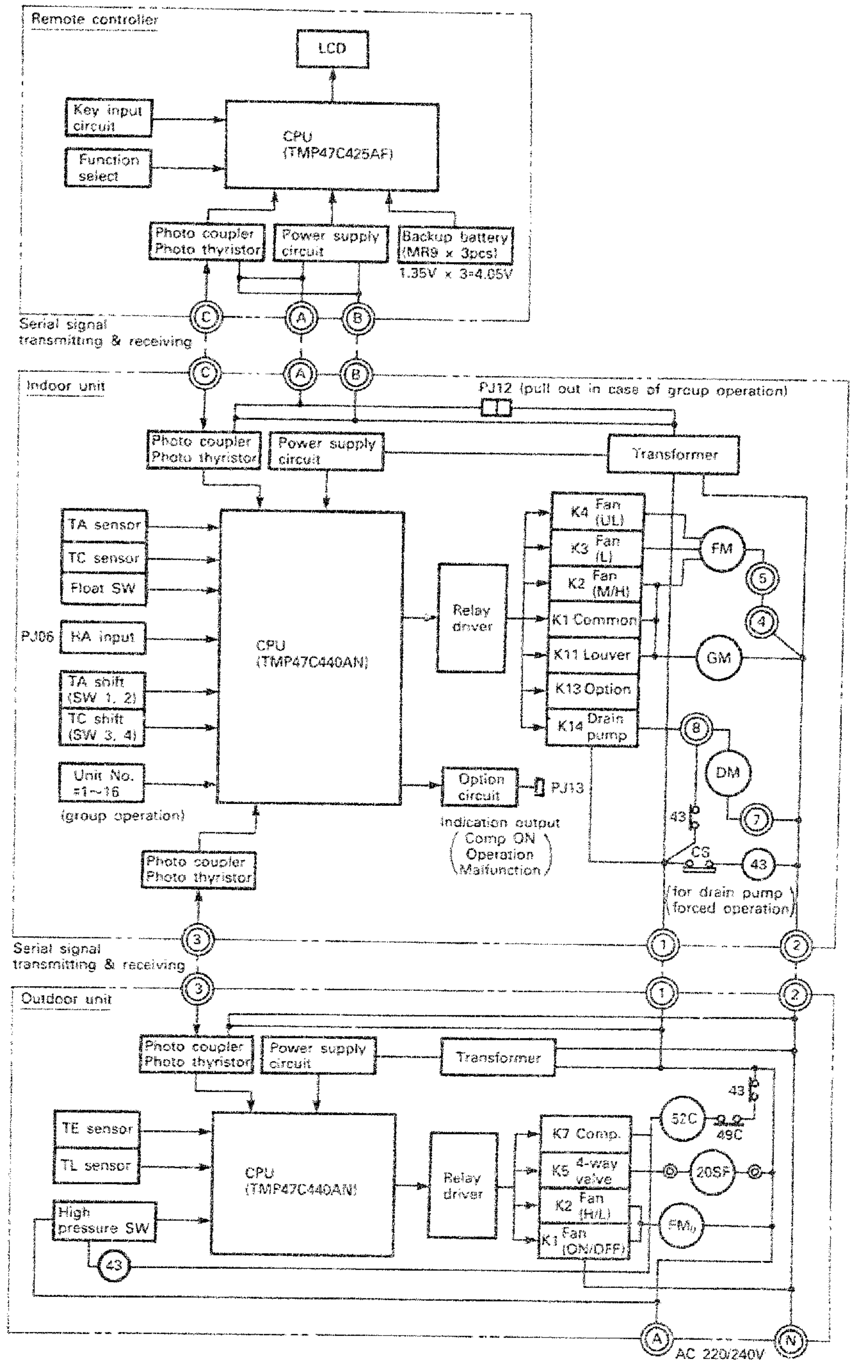
RAV-713UHE8 **RAV-S713HE8**



Line Pressure

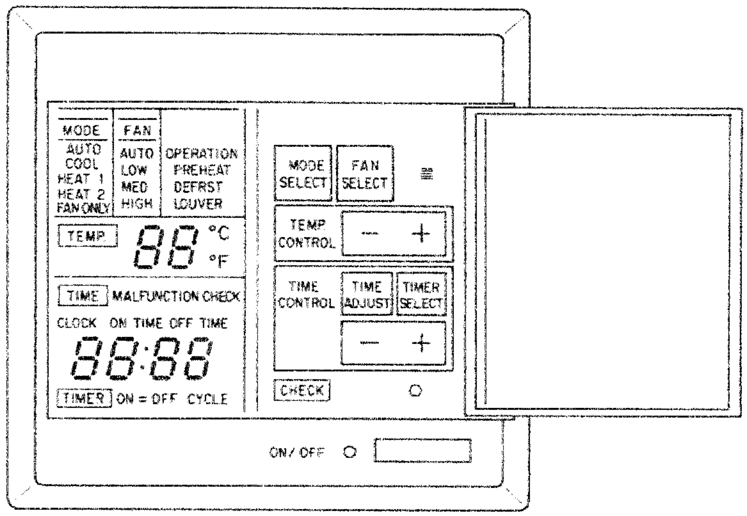
	Cooling	Heating
P_1	High pressure	Low pressure
P_2	Low pressure	High pressure
P_3	Medium pressure	Medium pressure

6. CONTROL CIRCUIT BLOCK DIAGRAM



7. REMOTE CONTROLLER

7.1 Remote controller



BUTTON	INDICATOR		OPERATION
ON/OFF		LED (RED)	Run/Stop
MODE SELECT	MODE	AUTO COOL HEAT 1 HEAT 2 FAN ONLY	Auto Changeover Cooling Heating Heating (with indoor fan operation at defrost) Fan only
FAN SELECT	FAN	AUTO LOW MED HIGH	Auto Fan Speed Control Low Fan Speed Med. Fan Speed High Fan Speed
		Louver	Auto Louver
TEMP. CONTROL	TEMP.	88 °C °F	Temperature setting
TIME ADJUST	TIME	(1) CLOCK (2) ON TIME (3) OFF TIME 88:88	(1) Present Time Adjust (2) ON Time Settng (3) OFF Time Setting
TIMER SELECT	TIMER	ON OFF ON → OFF ON ← OFF CYCLE	ON Timer OFF Timer ON → OFF Timer OFF ← ON Timer 24H Cycle Timer

7.2 Outline of remote controller's functions

NO.	KEY SWITCH	OUTLINE OF SPECIFICATIONS	REMARKS
1	[ON/OFF]	① When this button is pushed once, the air conditioner is turned on, with the operation lamp coming on. ② If pushed once more, it will be turned off, the operation lamp going off. ③ If pushed for 5 sec. in the mode of turning on the air conditioner, goes into test run mode.	Fan only after 30 min.
2	MODE SELECT	① Each time this button is pushed, the [MODE] setting is changed over cyclically, [AUTO] → [COOL] → [HEAT1] → [HEAT2] → [FAN ONLY] → [AUTO]. ② If pushed continuously, the setting will be changed in one step for every 0.5 sec.	
3	FAN SELECT	① Each time this button is pushed, the [FAN] setting is changed over cyclically, [AUTO] → [LOW] → [MED.] → [HIGH] → [AUTO]. ② If pushed continuously, the setting will be changed in one step for every 0.5 sec.	Fan speed
4	([LOUVER]) ≣	① When this button is pushed once, [LOUVER] indicator comes on. ② If pushed once more, [LOUVER] indicator will go off.	Provided for ceiling type
5	TEMP. CONTROL [+] [−]	① Each time [+] button is pushed, the [TEMP]. setting of temperature is raised by 1℃. ② If [+] is pushed continuously, the setting will be raised by 1℃ every 0.5 sec. ③ Each time [−] button is pushed, the setting of temperature is lowered by 1℃. ④ If [−] is pushed continuously, the setting will be lowered by 1℃ every 0.5 sec.	In the 18~29° CL range
6	TIME CONTROL TIME ADJUST [+] [−] TIMER SELECT	① Each time [TIME ADJUST] button is pushed, the [TIME] display is changed cyclically. The time can be changed while the TIME display stays flashing. [CLOCK] [12:00] → [CLOCK] (flashing) [12:00] → [ON TIME] (flashing) [6:00] → [OFF.TIME] (flashing) [18:00] ② While the TIME display stays flashing, the time gains one minute upon each pressing of [+] . ③ If [+] is pushed continuously, the time gains 10 minutes for every 0.25 sec. ④ While the TIME display stays flashing, the time goes back one minute upon each pressing of [−] . ⑤ If [−] is pushed continuously, the time goes back 10 minutes for every 0.25 sec. ⑥ Each time [TIMER SELECT] button is pushed, timer mode change over cyclically, [ ] (CONTINUE) → [ON] → [OFF] → [ON → OFF] → [ON ← OFF] → [CYCLE] → [ ]. ⑦ If pushed continuously, the timer mode will be changed in one step for every 0.5 sec.	If time is not set, 12:00 6:00 18:00 are set automatically. The digit of 1 min. becomes 0. The digit of 1 min. becomes 0.
7	CHECK	① Pressing this key for 0.5 sec. provides [MALFUNCTION CHECK], indicating on liquid crystal the contents of inspection in the sequence of (times of compressor-on) → (contents of malfunction for #1 unit) → (contents of mulfunction for #2 unit) →.... ② Pressing this key for 5 sec. gives "Indoor microcomputer reset mode" to reset the indoor microcomputer by way of hardware. ③ Pressing this key for 10 sec. gives "Check contents clear mode" to clear the contents of inspection stored in the remote controller provided, however, that times of compressor-on is not cleared.	The indication of the indoor unit which has not any malfunction content is skipped.
8	Reset	① By pressing the reset key, the remote controller is reset by way of hardware. (The setting/display are in initial values with the check memory cleared.)	

7.3 Timer operation

Contenuous operation and timer operations are available. The setting of timer operation can be done as follows: ON, OFF, ON → OFF, OFF → ON, ON ← → OFF CYCLE.

3-1) Time display

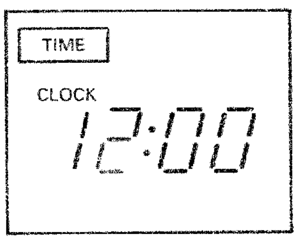
The present time is always displayed

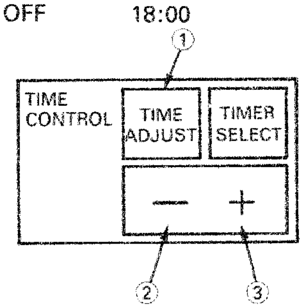
The display of the ON/OFF time is only in setting the time.
Once set, it will not changed even after carrying out the timer operation until the timer is reset.

Initial set time	The present time	12:00
	The time of ON	6:00
	The time of OFF	18:00

3-2) How to set the time

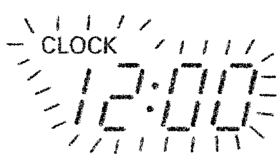
Liquid crystal





As to (−) and (+), change takes place by one minute by pressing once and 10 min./0.25 sec. by pressing continuously.

How to set the present time

1) 

[TIME ADJUST] switch is pressed.
[CLOCK] and Time figures flash.

2) Time is set by Time setting switch (−) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] three times gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time).

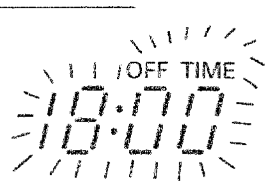
How to set ON TIME

1) 

[TIME ADJUST] switch is pressed twice.
[ON TIME] and Time figures flash.

2) Time is set by Time setting switch (−) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] twice gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time)

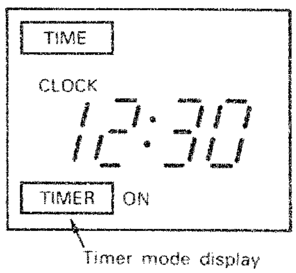
How to set OFF TIME

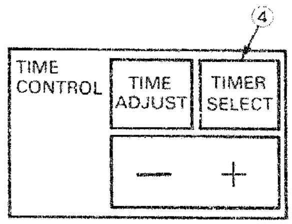
1) 

[TIME ADJUST] switch is pressed three times.
[OFF TIME] and Time figures flash.

2) Time is set by Time setting switch (−) or (+). The setting is finished when releasing. Pressing [TIME ADJUST] once gives the display of the present time.
(If left as it is, after 15 sec. the display will go back to the present time)

3-3) How to set the timer operation





The following can be chosen sequentially by pressing [TIMER SELECT] switch:

1) [TIMER] ON
2) [TIMER] OFF
3) [TIMER] ON → OFF
4) [TIMER] ON ← OFF
5) [TIMER] CYCLE

* Be sure to set the present time.
* In case of reoperating after finishing timer operation, if [TIMER SELECT] is not altered, the timer operation will be performed again.

Timer ON operation

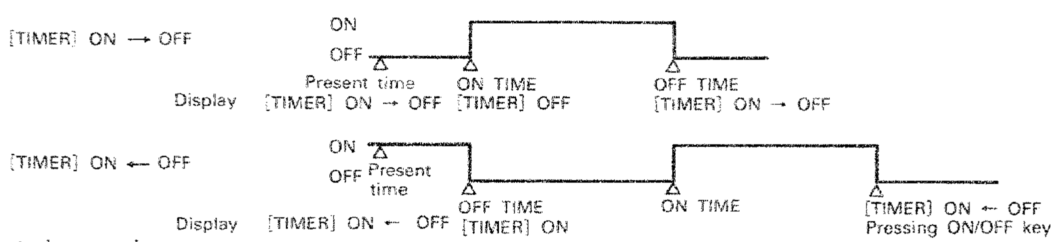
1) [TIMER] ON is applied.
2) ON/OFF key is pressed. Then LED is lighted.
   When the set [ON TIME] comes, the operation starts and OPERATION display comes on the liquid crystal, and the [TIMER] ON display goes off.
3) LED and OPERATION display goes off upon pressing ON/OFF key for stopping and [TIMER] ON is displayed.

Timer OFF operation

1) [TIMER] OFF is applied.
2) ON/OFF key is pressed. Then LED is lighted and the operation starts with OPERATION displayed on the liquid crystal.
3) When the set [OFF TIME] comes, the operation stops and the LED, OPERATION display goes off with [TIMER] OFF displayed.

ON ←→ OFF timer operation

1) [TIMER] ON → OFF or [TIMER] ON ← OFF is applied.
2) ON/OFF key is pressed. LED comes on and the operation is performed as below:

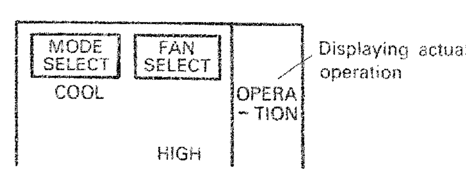


Repeated operation

1) [TIMER] CYCLE is applied.
2) ON/OFF key is pressed. Then LED is lighted and ON ←→ OFF timer operation is repeated according to the ON time and OFF time (repeating every day as it is a 24-hour timer).
3) The operation key is pressed. LED goes off and operation stops.

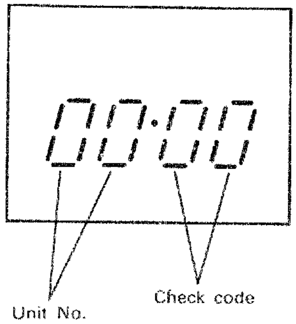
Timer stand-by display and operation display

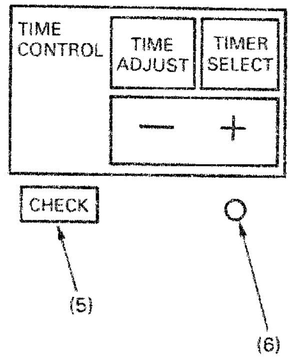
Waiting on the timer is displayed by LED lighting while the actual operation is displayed on OPERATION on liquid crystal.



7.4 Malfunction check monitor

4-1) The times of thermostat ON as well as the check code are displayed on the time display area by pressing CHECK key.





(5) Check key
Provides check code display by pressing for one second and indoor microcomputer reset by pressing for 5 seconds.

* Remote controller clear by pressing the key for 10 sec. Check code is cleared (normally not used).

(6) Reset key (pushed by a needle and the like) Resetting remote controller (to the initial setting)

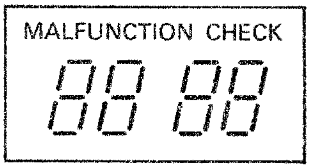
Judgement from operation status

	OPERATION STATUS	CODE	CAUSE
1.	Thermostat stays off in cooling: it is not turned off in heating	0C	Open-circuit in room temperature sensor.
	Thermostat stays off in heating: it is not turned off in cooling		Short-circuit in room temperature sensor.
2.	Indoor fan stays off in heating.	0d	Open-circuit in indoor heat-exchanger sensor.
	Outdoor fan continues ON-OFF operation in heating.		Short-circuit in indoor heat-exchanger sensor.
3.	While indoor unit is in operation, outdoor unit keeps on stoppage Neon lamp comes on.	0b	Abnormality in drain system. Fault of drain pump. Drain pipe clogged.
4.	Though indoor unit operates, outdoor unit keeps on stopping.	04	Abnormality in connecting cable between indoor and outdoor units.
5.	Indoor fan does not work in heating operation. Warm air comes out in cooling operation.	08	4-way valve coil burnt out, pipe clogged, abnormality in indoor heat-exchanger sensor.
6.	Indoor fan at LOW speed in cooling operation with the outdoor remaining in stoppage.	09	Refrigerant gas in shortage. Abnormality in indoor heat-exchanger sensor.
7.	Full stop	18	Open or short-circuit in outdoor TE sensor.
8.	Full stop	19	Open or short-circuit in outdoor TL sensor.
9.	Full stop	21	Pressure switch does not reset within the set time.
10.	Full stop	1C	Pressure switch, overcurrent sensor operated.

4-2) How to read malfunction check monitor display

By pressing [CHECK] key, times of No. 1 unit compressor-ON actuations as well as the check code information of 2 faults x 16 units are displyed on the time display area. (2 sec. per one phenomenon)

⟨Times of compressor-ON⟩

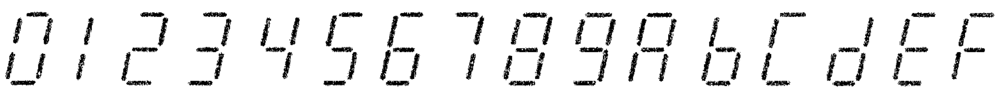


Display in 4 digits of hex-adecial notation

Ex. In case of the number of times of compressor actuations of 164.

$16^3 \times 0 + 16^2 \times 0 + 16 \times 10 + 4 = 164$

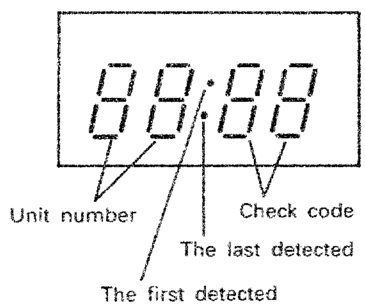
Display in 7 segments



hexadecimal digit

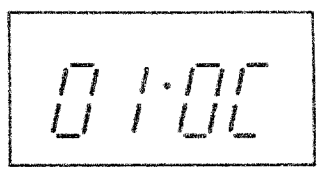
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 decimal digit

⟨Check code information⟩

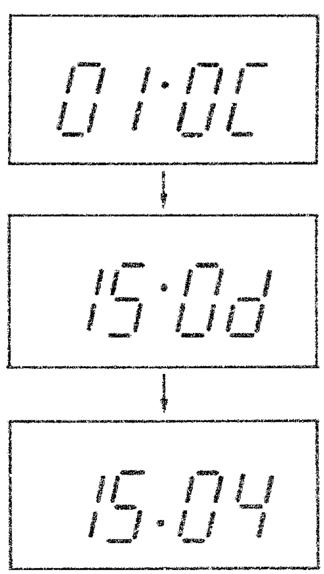


No display is made if there is no fault.

Ex. In case of room temperature sensor of No.1 unit in trouble.



For No. 15 unit, firstly room temperature sensor and secondaly connecting cable between the indoor and outdoor units (serial signal) are in trouble.



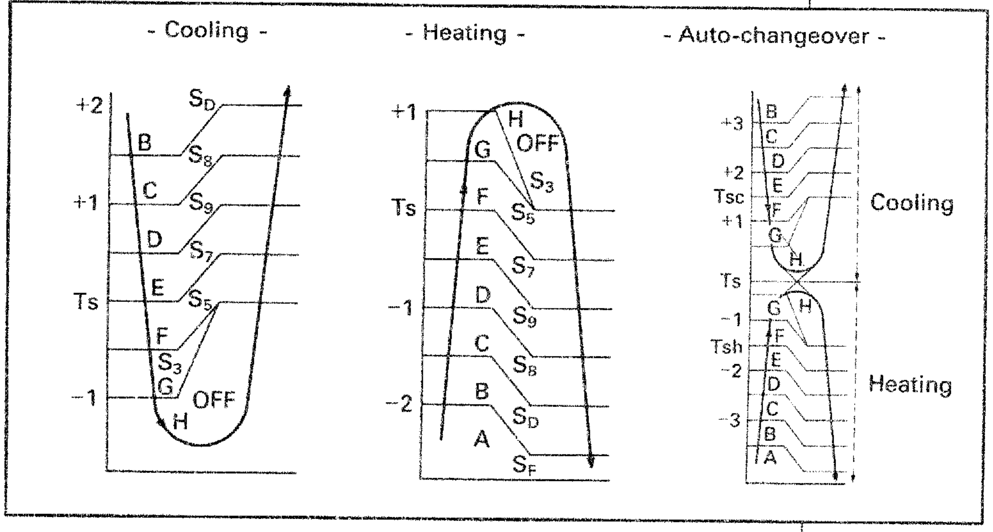
4-3) List of Check Code

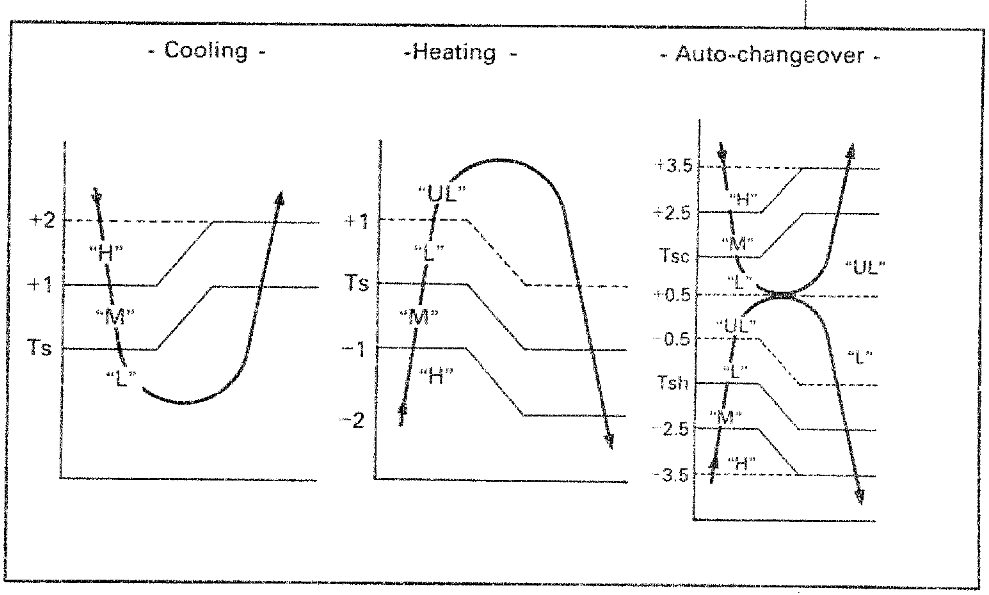
DIAGNOSTIC FUNCTIONS			JUDGEMENT AND ACTION
CHECK CODE	SYMPTOM	STATUS OF AIR CONDITIONER	
0C	ROOM TEMP. SENSOR (TA). Out of place, break, short-circuit.	Operation continuing	1. Check for indoor temp. sensor. 2. Check for indoor PC board.
0d	INDOOR HEAT-EXCHANGER SENSOR (TC). Out of place, break, short-circuit.	Operation continuing	1. Check for indoor heat-exchanger sensor. 2. Check for indoor PC board.
0b	FLOAT SWITCH float circuit out of position, break.	Outdoor unit stops	1. Fault in drain pump. 2. Drain pipe clogged. 3. Check for indoor PC board.
04	RETURN SIGNAL NOT COMING TO INDOOR 1) Wrong wiring in connecting cable (serial signal).	Operation continuing	1. If outdoor unit does not work at all. (1) Check for connecting cable correct wrong wiring. (2) Check for outdoor PC board. 2. If operates normally. Between indoor terminal plates 2 and 3, return signal is : Available: Check for indoor PC board. Not available: Check for outdoor PC board.
08	4-WAY VALVE SYSTEM 1) Indoor heat-exchanger temperature rises, after starting cooling operation. 2) Indoor heat-exchanger temperature drops after starting heating operation.	Operation continuing	1. Check for 4-way valve. 2. Check for 2-way valve and check valve. 3. Wrong with indoor heat exchanger sensor. 4. Check for indoor PC board.
09	OTHER CYCLE SYSTEM 1) Indoor heat exchange temperature does change after starting cooling/heating operation.	Operation continuing	1. Compressor case thermostat, IOL operation. (contactor OFF, compressor stops) 2. Indoor heat-exchange sensor out of place. 3. Check for indoor PC board.
	2) When transmitting instruction for stopping compressor by freeze preventing control.	Outdoor unit stops (indoor fan L)	1. Check for charged amount of refrigerant gas. (Gas shortage → gas supplement, check for gas leaks) 2. Indoor fan locked.

18	DEFROST SENSOR (TE) Out of place, break, short-circuit.	Full stop	1. Check for defrosting sensor. 2. Check for outdoor PC board.
19	OUTDOOR HEAT EXCHANGER SENSOR (TL) Out of place, break, short-circuit.	Full stop	1. Check for outdoor heat exchange sensor. 2. Check for outdoor PC board.
21	HIGH PRESSURE SWITCH High pressure switch does not reset. (5 sec : in cooling 30 sec : in heating)	Full stop	1. Check for high pressure switch. 2. Check for outdoor PC board.
1C	OTHER ABNORMALITY OF OUTDOOR UNIT Compressor dose not operate. Start once, but soon after stop by OCR	Full stop	1. Check for compressor. 2. Check for wiring of compressor. (lack of phase, short circuit) 3. Check for voltage. 4. Check for outdoor PC board.

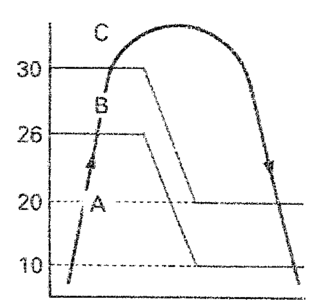
8. OUTLINE OF CONTROL CIRCUIT

<table>
<tr><th>NO.</th><th>ITEM</th><th>OUTLINE OF SPECIFICATIONS</th><th>REMARKS</th></tr>
<tr><td>1</td><td>Discrimination</td><td>Discrimination of outdoor unit is performed either in the reset of power source or when stopping from operating condition, and the controlling is changed over in accordance with the result of discrimination. [Heating S1] is used as the discriminating signal, [Inverter heat pump], [Normal heat pump], and [Cooling only] being discriminated according to the contents of reverse signal from the outdoor unit.</td><td></td></tr>
<tr><td>2</td><td>Operation change-over</td><td>Operation mode is changed over according to operation mode select instruction from the remote controller.
<table>
<tr><th>REMOTE CONTROLLER INSTRUNCTION</th><th>OUTLINE OF CONTROL</th></tr>
<tr><td>Stop</td><td>Stopping air conditioner</td></tr>
<tr><td>Auto</td><td>Performing automatic change-over</td></tr>
<tr><td>Cool</td><td>Performing cooling operation</td></tr>
<tr><td>Heat 1</td><td>Performing heating operation</td></tr>
<tr><td>Heat 2</td><td>Performing heating operation with indoor fan operation at defrosting</td></tr>
<tr><td>Fan only</td><td>Performing fan only operation</td></tr>
</table>
</td><td></td></tr>
<tr><td>3</td><td>Controlling room temperature</td><td>3-1 Adjusting range
<table>
<tr><th></th><th>In cooling</th><th>In heating</th></tr>
<tr><td>Remote controller setting tempera-ture</td><td>18 ~ 29</td><td>18 ~ 29</td></tr>
<tr><td>Operating temperature</td><td>18 ~ 29</td><td>20 ~ 31</td></tr>
</table>
3-2 Operating point is compressor - off.
3-3 Operating temperature accuracy: ±1°C
3-4 Differential: 1 deg</td><td></td></tr>
<tr><td></td><td>Correcting temperature compensation</td><td>3-5 Room temperature controlled in the heating operation can be corrected by dip switch of indoor microcomputer.
<table>
<tr><td>Dip switch 1
Setting 2</td><td>ON
ON</td><td>ON
OFF</td><td>OFF
ON</td><td>OFF
OFF</td></tr>
<tr><td>Control temperature compensation</td><td>0deg</td><td>2deg</td><td>4deg</td><td>6deg</td></tr>
</table>
</td><td>Ts(Max)=35°C</td></tr>
<tr><td>4</td><td>Automatic capacity control (Thermo-control)</td><td>4-1 Instruction of operation frequency is given to outdoor unit based on the table below according to the difference between Ta and Ts.
4-2 Compressor is turned on at the frequecny instruction of [S3] or higher, and turned off below [S3].</td><td></td></tr>
</table>

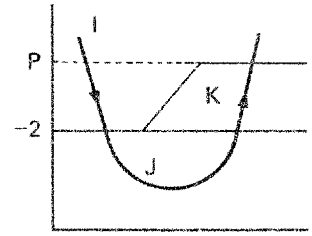
<table>
<tr><th>NO.</th><th>ITEM</th><th>OUTLINE OF SPECIFICATIONS</th><th>REMARKS</th></tr>
<tr><td></td><td></td><td>


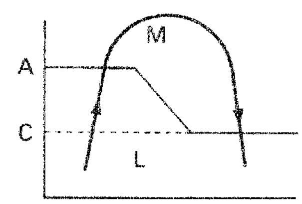
</td><td></td></tr>
<tr><td>5</td><td>Fan speed control</td><td>5-1 [HIGH], [MED.], [LOW] and [AUTO] are available.
5-2 [Ultra low] or [Stop] when thermostat is turned off while heating is being performed.
5-3 In the auto fan, the fan speed is changed by the difference between Ta and Ts, as shown below.


</td><td>[Stop] is cold draft prevention by Tc.</td></tr>
<tr><td>6</td><td>Cold draft preventing control</td><td>When performing heating operation, indoor fan control is carried out as follows based on temperature detection of Tc sensor.

C zone: Depending upon fan speed setting of the remote controller
B zone: Indoor fan at "L"
A zone: Fan stop


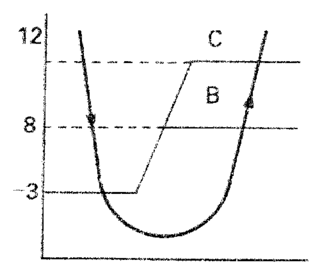
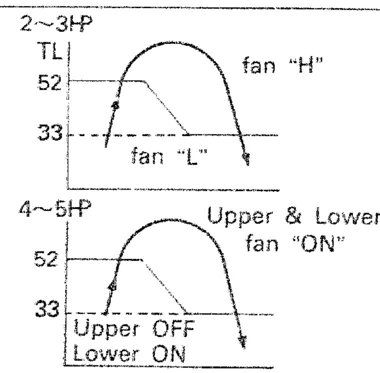
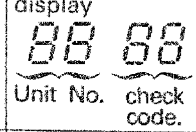
</td><td></td></tr>
</table>

<table>
<tr><th>NO.</th><th>ITEM</th><th>OUTLINE OF SPECIFICATIONS</th><th>REMARKS</th></tr>
<tr><td>7</td><td>Freeze preventing control (Low temp. release)</td><td>When performing cooling operation, the following control is done based on temperature detection of Tc sensor.


① When starting the operation, the point P is made +3℃.
② Frequency instructed to the outdoor unit is reduced when [J] zone is detected for 1 minute continuously.
③ Frequency instructed to the outdoor unit is reduced each time 2 minutes have elapsed thereafter.
④ When [K] zone is detected, timer counting is discontinued and held on.
⑤ When [I] zone is detected, timer is cleared for returning back to ordinary operation.
⑥ When frequency instructed to the outdoor unit has become H, the point P is changed to +12℃ to be covered by check display. When [I] zone is reached, the temperature is returned back to +3℃.</td><td>H=OFF</td></tr>
<tr><td>8</td><td>High temperature release control</td><td>When performing heating operation, the following control is done based on temperature detection of Tc sensor.
① In [M] zone, release signal is transmitted. Outdoor fan is turned off at the shortest for 3 minutes based on this signal.
② The control point for A and C can be chosen from the below table:


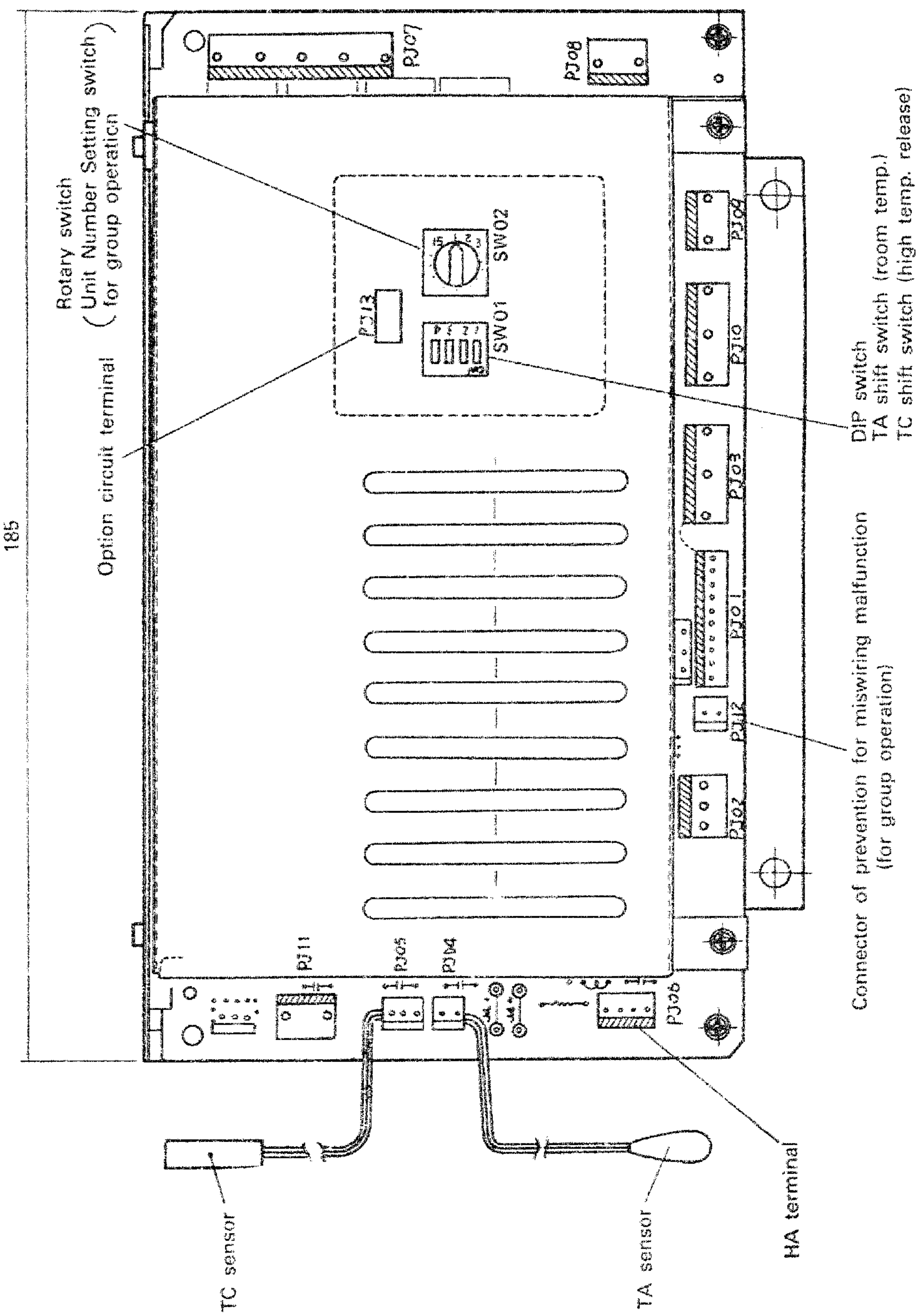
<table>
<tr><td>Dip switch Setting</td><td>3
4</td><td>ON
ON</td><td>ON
OFF</td><td>OFF
ON</td><td>OFF
OFF</td></tr>
<tr><td></td><td>A/C</td><td>54/52</td><td>58/56</td><td>60/58</td><td>–</td></tr>
</table></td><td>Interval operation of outdoor upper fan at low is done in the outdoor unit select B mode. (Outdoor fan)

B=4HP ,5HP</td></tr>
<tr><td>9</td><td>Drain pump control</td><td>9-1 When [Cooling] operation is performed, drain pump is actuated.
9-2 While overflow switch is operated, compressor is turned off and drain pump works.
9-3 When overflow switch is operated for a period of 2 minutes, it will become the subject of check display.</td><td>Provided for cassette type</td></tr>
<tr><td>10</td><td>Residual heat removal</td><td>When stoppage takes place in [HEAT 2] operation, indoor fan is operated in [LOW] for 30 sec.</td><td></td></tr>
<tr><td>11</td><td>Auto louver control</td><td>When receiving louver signal from remote controller, auto louver operation is performed if indoor fan is being in operation.</td><td>Provided for ceiling type</td></tr>
<tr><td>12</td><td>HA control</td><td>The air conditioner can be turned on and off by pulse signal DC5V.</td><td>External signal</td></tr>
<tr><td>13</td><td>Test operation</td><td>13-1 If Remote controller's ON/OFF switch is pressed 5 minutes continuously, the unit goes into test run mode, and fixed-frequency operation is performed with the indoor fan in the [HIGH].
13-2 After continuing the operation for 30 minutes, [Fan only] operation is initiated.</td><td>Instructed frequency [S7]</td></tr>
</table>

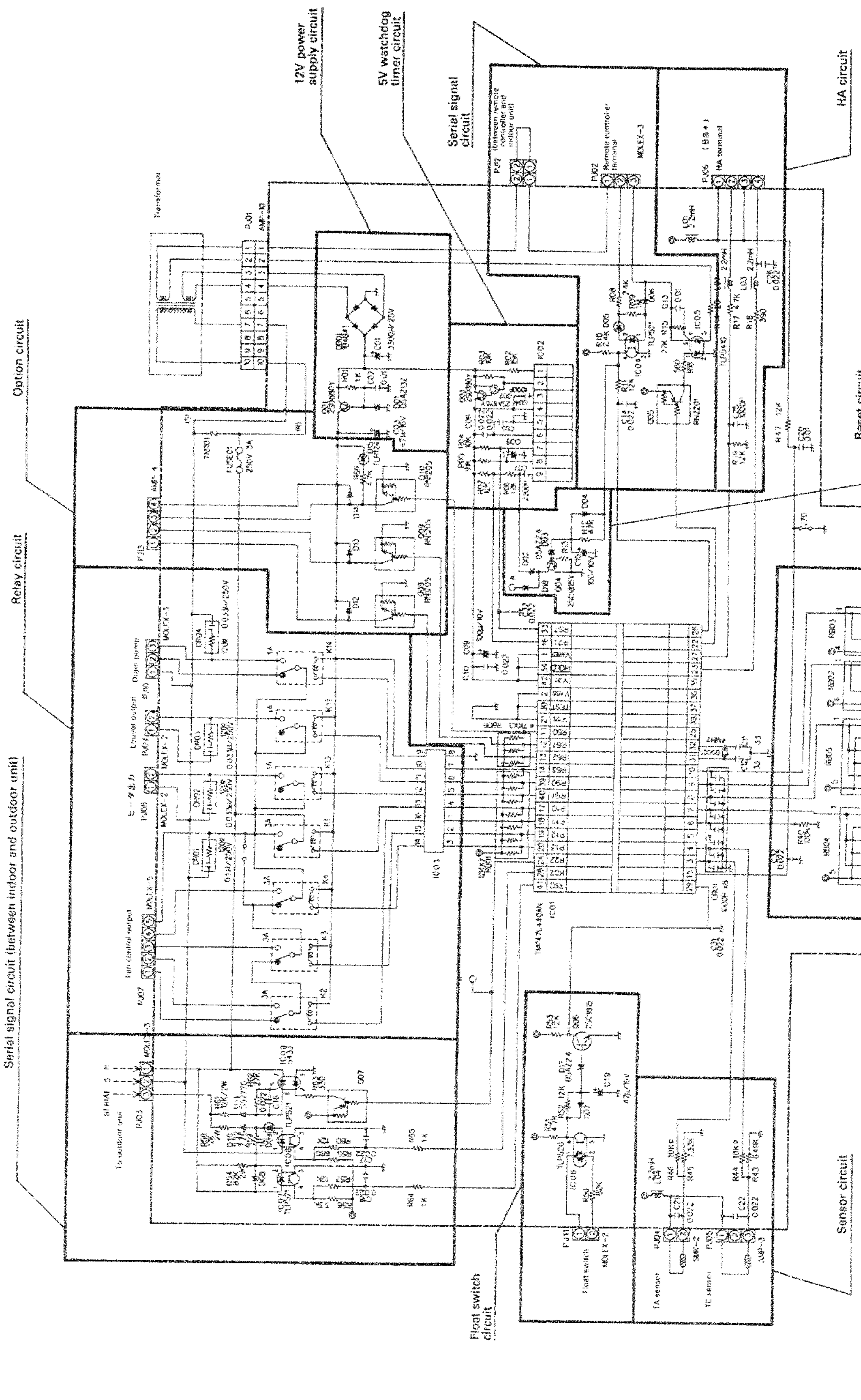
NO.	ITEM	OUTLINE OF SPECIFICATIONS	REMARKS
14	High pressure release	The following control is performed when high pressure switch of the outdoor unit is actuated. ① In cooling operation Compressor is turned off and if the high pressure switch does not reset for 5 seconds continuously thereafter, it is judged abnormal. ② In heating operation Compressor is turned off and if the high pressure switch does not reset for 30 seconds continuously thereafter,it is judged abnormal. If the switch resets within 30 sec., the compressor restarts 2 minutes and 30 sec. later. And if this process is repeated, the release by outdoor fan and shift of compressor-on point will be done. ③ In defrosting operation Compressor is turned off, the operation returning back to heating operation.	〈Outdoor unit control〉 LED lamp comes on in abnormal condition, being abnormal code transmitted to indoor unit.
15	Defrosting	15-1 In heating operation, defrosting is made based on outdoor heat exchange temperature Te. 15-2 When cumulative working time of the compressor in [A] zone has amounted to 55 minutes, defrosting operation starts. (25 minutes initially) 15-3 The longest defrosting time is 12 minutes, 60 sec. in the case of turning into [B] zone, and immediate returning back when [C] zone is reached. 	〈Outdoor unit control〉
16	Low ambient cooling	16-1 Control on outdoor fan is made to meet with cooling at low outdoor temperature based on outdoor heat exchange temperature TL. 16-2 Control by outdoor heat exchange temperature TL is illustrated in the right. 	〈Outdoor unit control〉
17	Check display	Fault diagnosis is carried out by check for serial signal transmission and reception with outdoor unit as well as the self check by indoor microcomputer. And check code is transmitted to protective operation/remote controller based on the contents of it. In the remote controller, check code is displayed on the liquid crystal by pressing [CHECK] key.	See other item: Using [TIME] display 
18	Anti-restart timer	The outdoor unit delays restarting for 2.5 min. to prevent short cycling compressor operation.	
19	Group operation control	Up to 16 units can be controlled in same setting condition by one remote controller. However, thermo-control function is independent. Respective delayed start time for preventing simultaneous large starting current can be by different setting of the unit No. switch on the indoor PC boad.	Refer to P.38

9. DESCRIPTION OF INDOOR UNIT CONTROL CIRCUIT

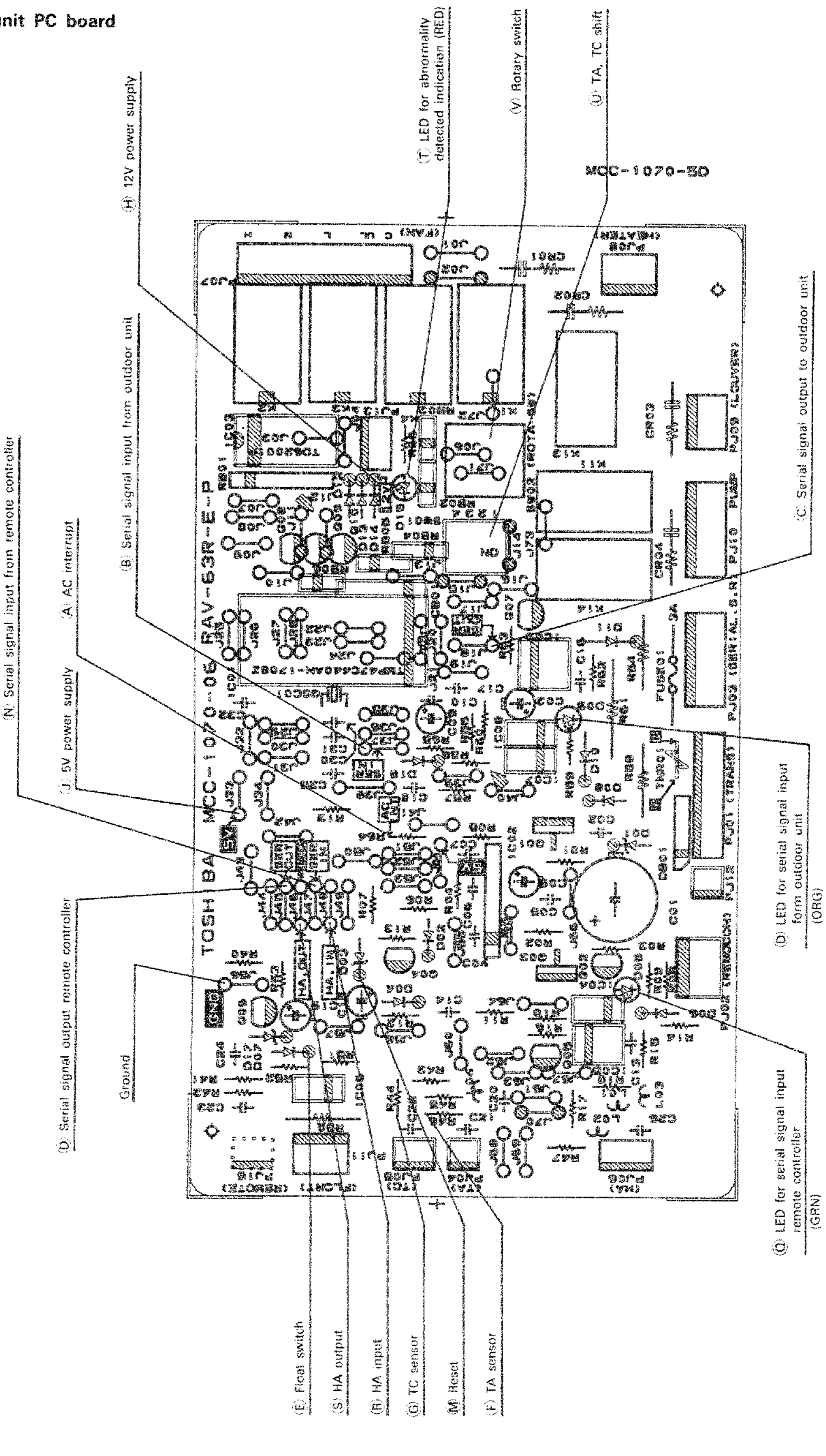
9.1 Indoor unit control box



9.2 Indoor unit control circuit diagram



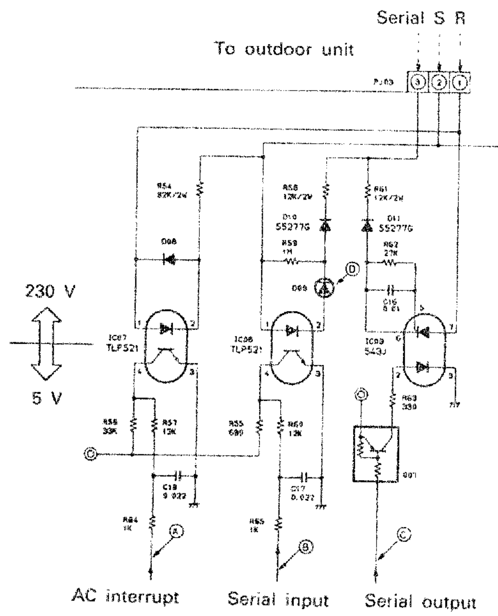
9.3 Indoor unit PC board

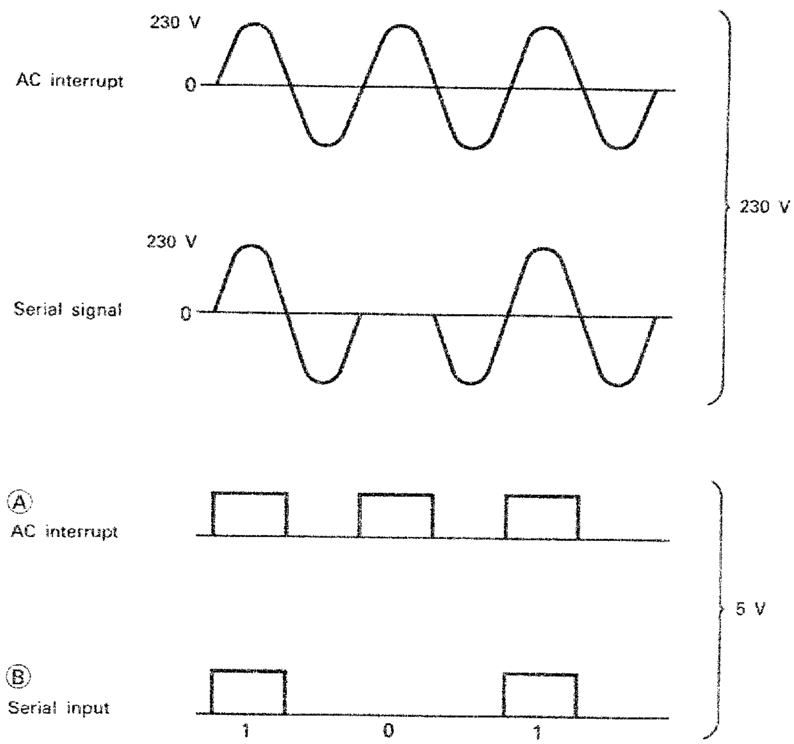


Serial signal circuit (between outdoor and indoor units)

This is a circuit for transmitting and receiving the signals between the indoor and outdoor units in serial signal. As 230V is used for transmitting the signal, the microcomputer section is insulated by means of photo-coupler with the voltage reduced to 5V.

With AC interrupt, judgement is made as to the presence or absence of serial signal based on the reference pulse taken out from the voltage across R and S.

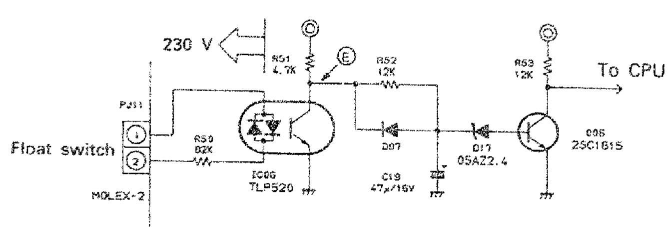




Ⓐ / Ⓑ ... are measurement points on the printed circuit board.
Ⓓ provides flashing (orange) on LED in the serial input.

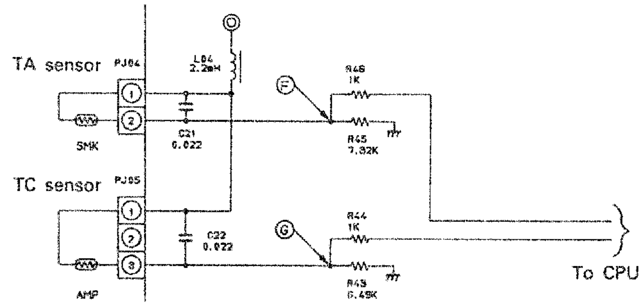
Float switch circuit

In normal condition in which float switch is not operated, 230V is applied across the pins 1 and 2. At this time, point Ⓔ is at the GND level. If the float switch is operated, Ⓔ will be at the level of 5V.



Sensor circuit

This circuit detects the temperature by dividing voltage with resistance and sensor and bringing the voltage value into CPU, using the characteristics of the sensor that resistance varies with different temperatures. TA and TC have the same circuit composition.



When TA and TC are at 25°C approximately, the voltage level is some 2V both at points Ⓕ and Ⓖ.
If Ⓕ / Ⓖ are at GND or 5V, abnormal condition prevails such as opening or short-circuit of the sensor.

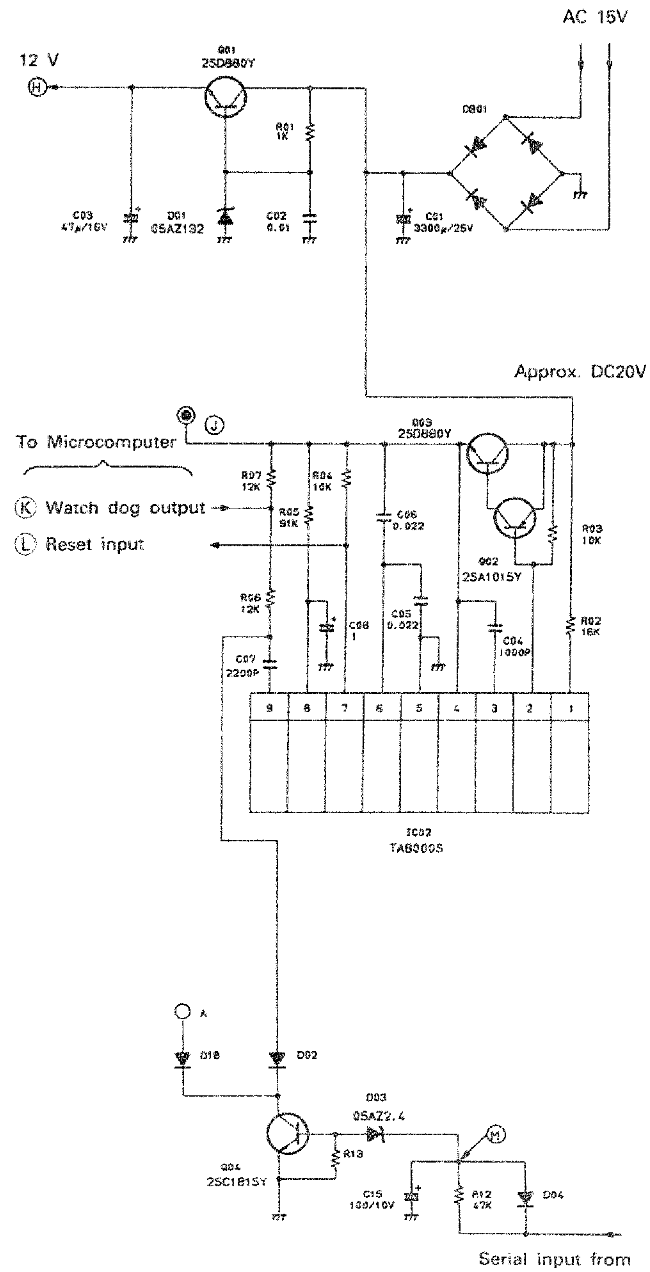
12V power source circuit

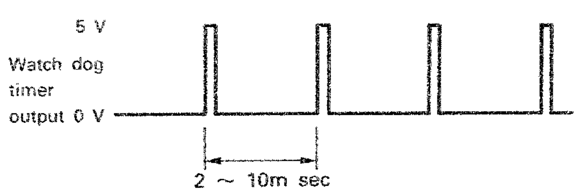
Full-wave rectification by diode bridge (DB01) of alternate current supplied from power transformer followed by the provision of transistor (Q01) gives DC12V power source (Ⓗ).

5V watchdog timer circuit

Built-in IC (TA8000S) is used to produce 5V power source (Ⓙ). Also, it sends signals to reset port (Ⓛ) of microcomputer which is in stand-by at 0V and starts its operation with the signal of 5V.

Watchdog timer output gives the signal from microcomputer as illustrated below. This indicates that the microcomputer is working in normal routine. For example if the microcomputer is straying due to noise and so on, this waveform is not produced. In case there is no waveform, it plays the role of restoring normal condition by inputting the resetting "0V" to the microcomputer.



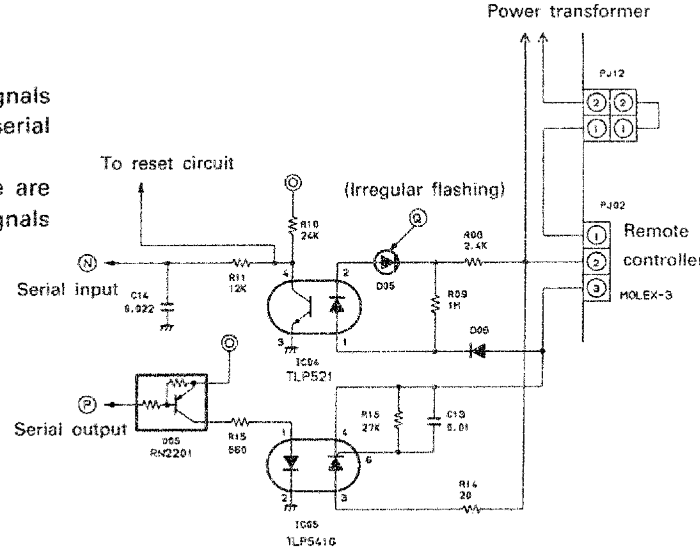


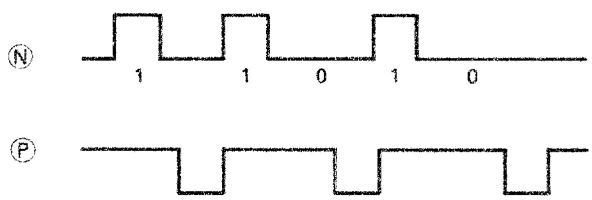
Reset circuit

This circuit makes indoor microcomputer reset by way of hardware when you keeps on pressing the check key of remote controller for longer than a predetermined period. It plays the role of resetting microcomputer from the remote controller when it strays. The point (Ⓜ), which is normally at the level of 5V, drops down to the GND level in the reset operation.

Serial signal circuit
(Between remote controller and indoor unit)

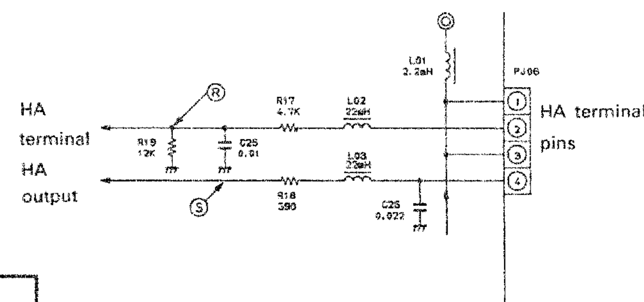
This is the circuit for transmitting and receiving the signals between the remote controller and indoor unit in serial signal.
Point Ⓠ is a LED (green) which flashes when there are signals from the remote controller. At Ⓝ and Ⓟ, the signals as illustrated below are output.

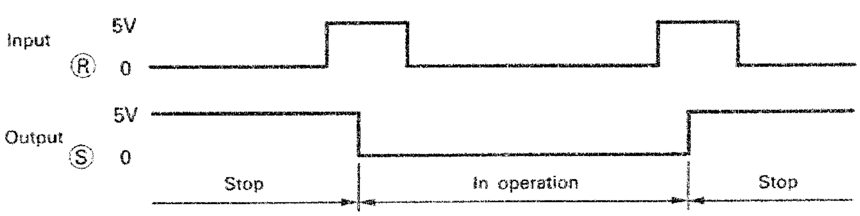




HA circuit

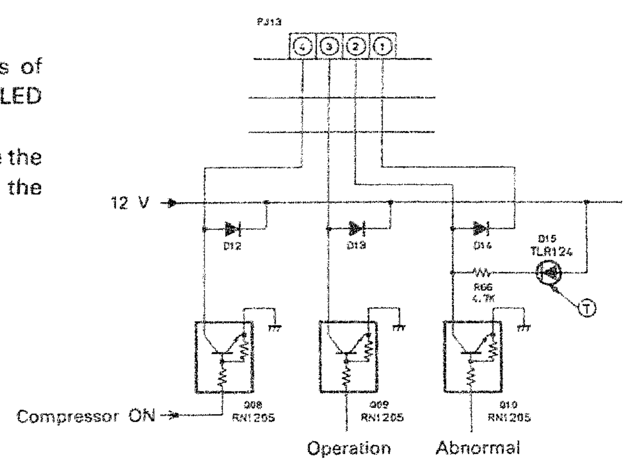
This is a circuit for turning the air conditioner on and off externally. The connector pins 1 and 3 are at 5V. Pin 2, which is signal input, performs on-and-off operation each time 5V signals comes. HA output is at GND in operation and at 5V in stoppage.

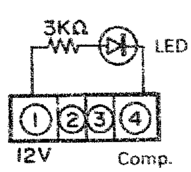




Optional circuit

A circuit which allows for the take-out of the signals of abnormal, operation and Compressor-ON. Point Ⓣ is a LED which lights at abnormal.
The connector pin 1 outputs 12V. When you want to see the signal of compressor-ON, you can do it simply with the circuit below.

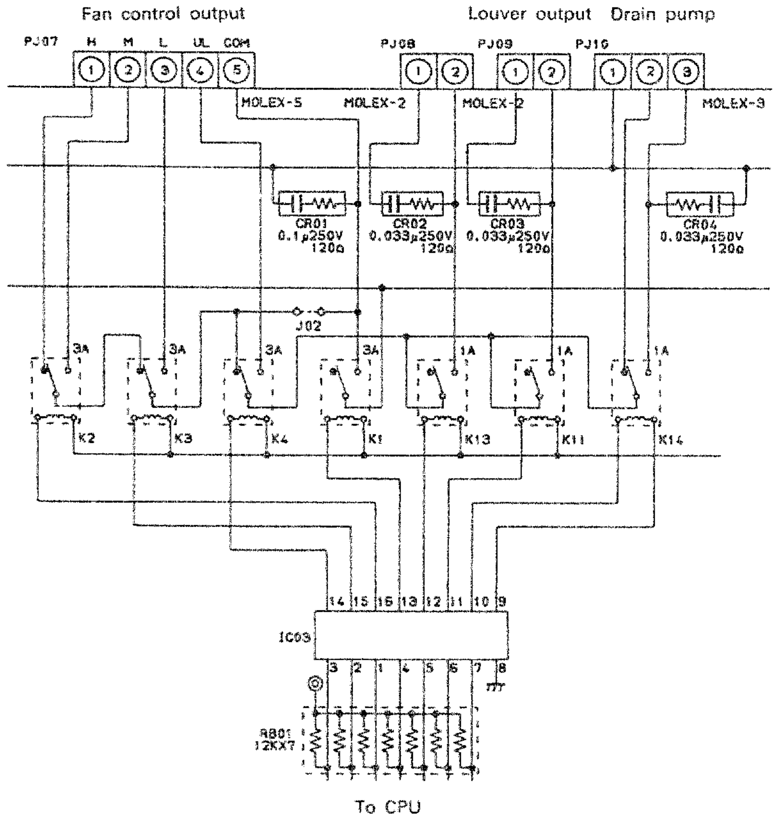




Relay circuit

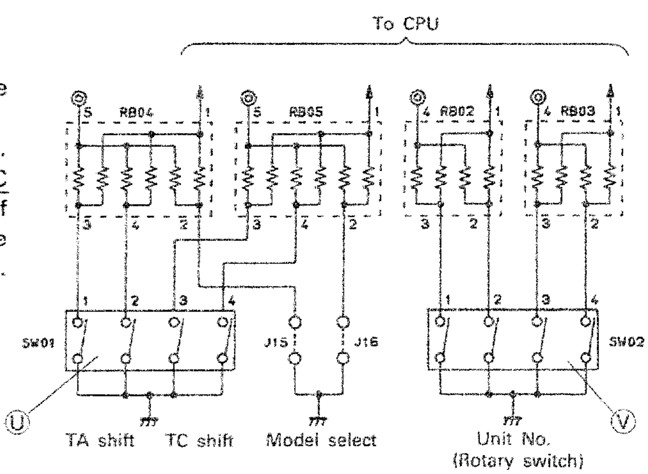
The relay circuit consists of the diagram in the righthand side.
The relay performs the following functions:

K1 : Turning fan on and off
K2 : Changing over H/M of fan
K3 : L tap of fan
K4 : UL tap of fan
K11: Turning louver on and off
K14: Turning drain pump on and off (① - ③)



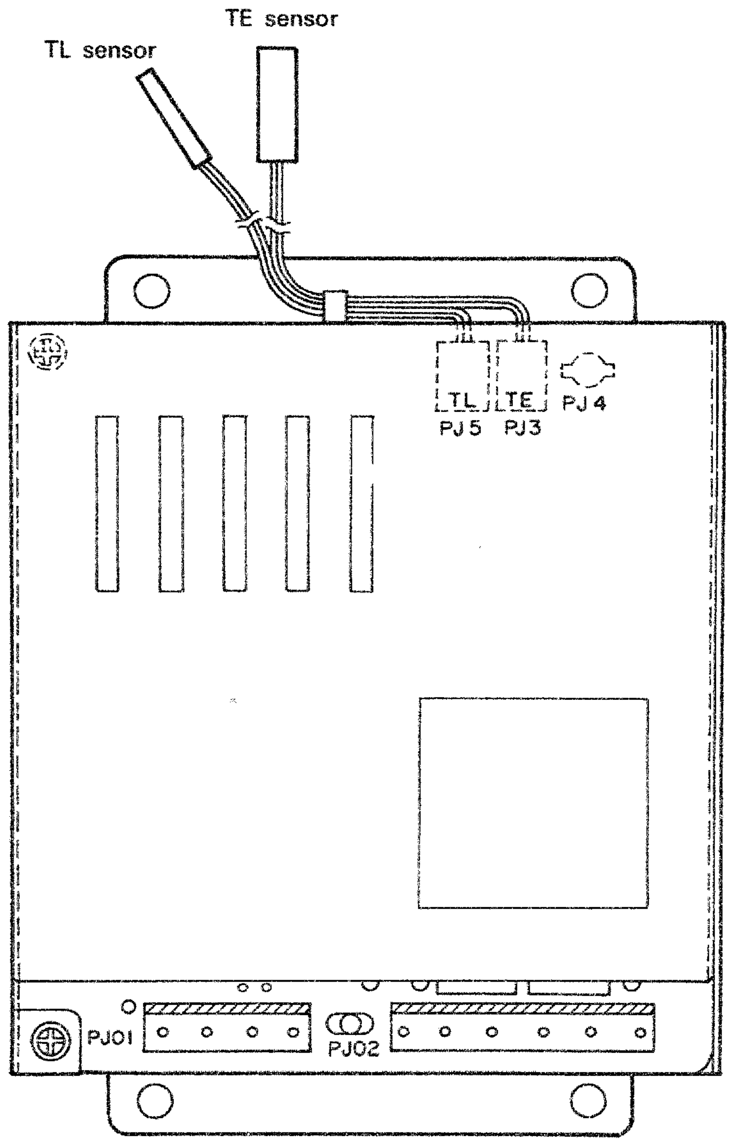
Switch circuit

TA shift, TC shift and unit No. are changed over by the switch.
TA shift and TC shift are set in factory with unit No. at "1". In servicing, the setting should be made to the same TA/TC shift as the PC board attached originally. In case of operating one single unit, unit No "1" will do. With the operation of many units (multi units control) the unit No. should be adjusted in such a way as 1, 2, 3.....

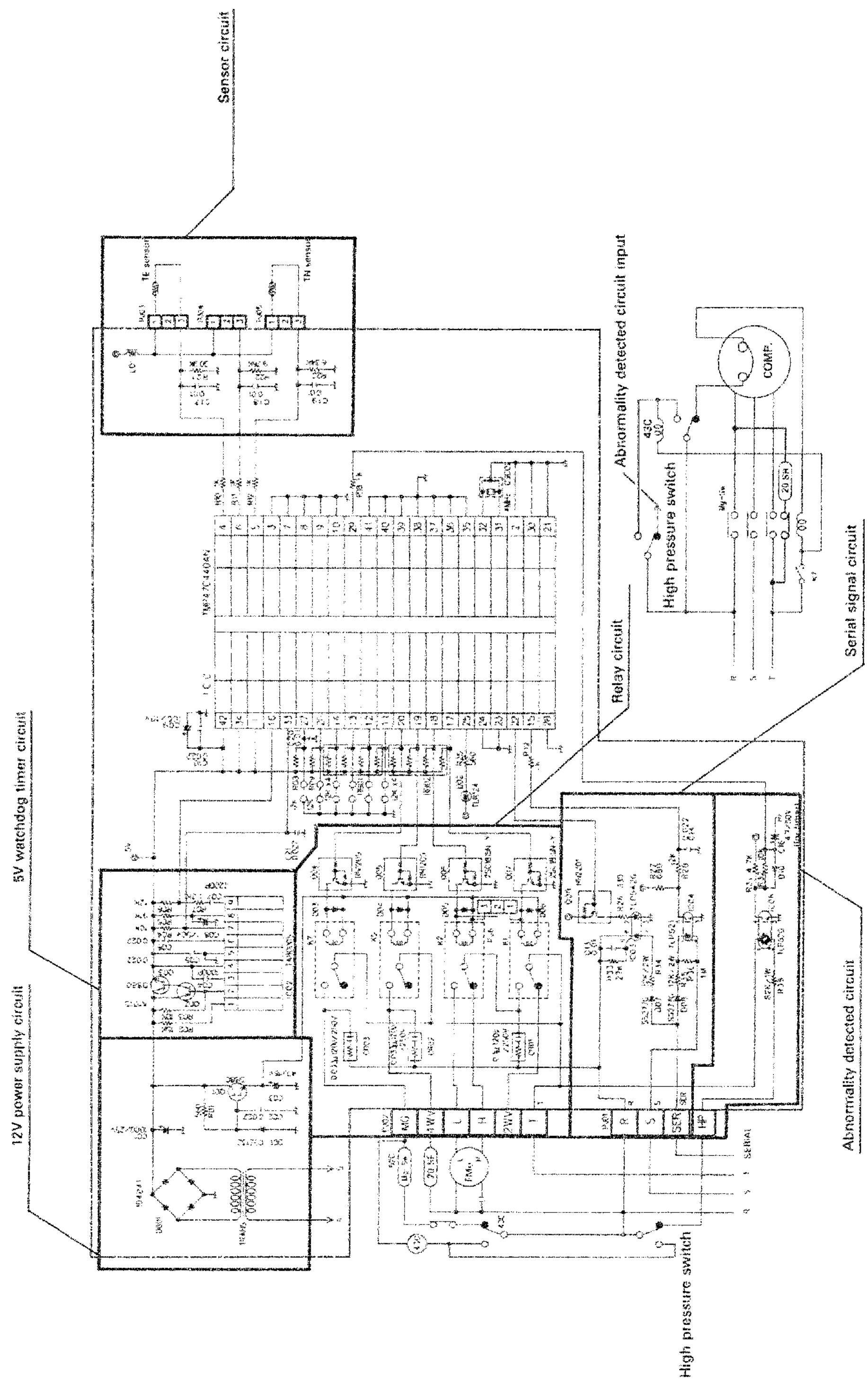


10. DESCRIPTION OF OUTDOOR UNIT CONTROL CIRCUIT

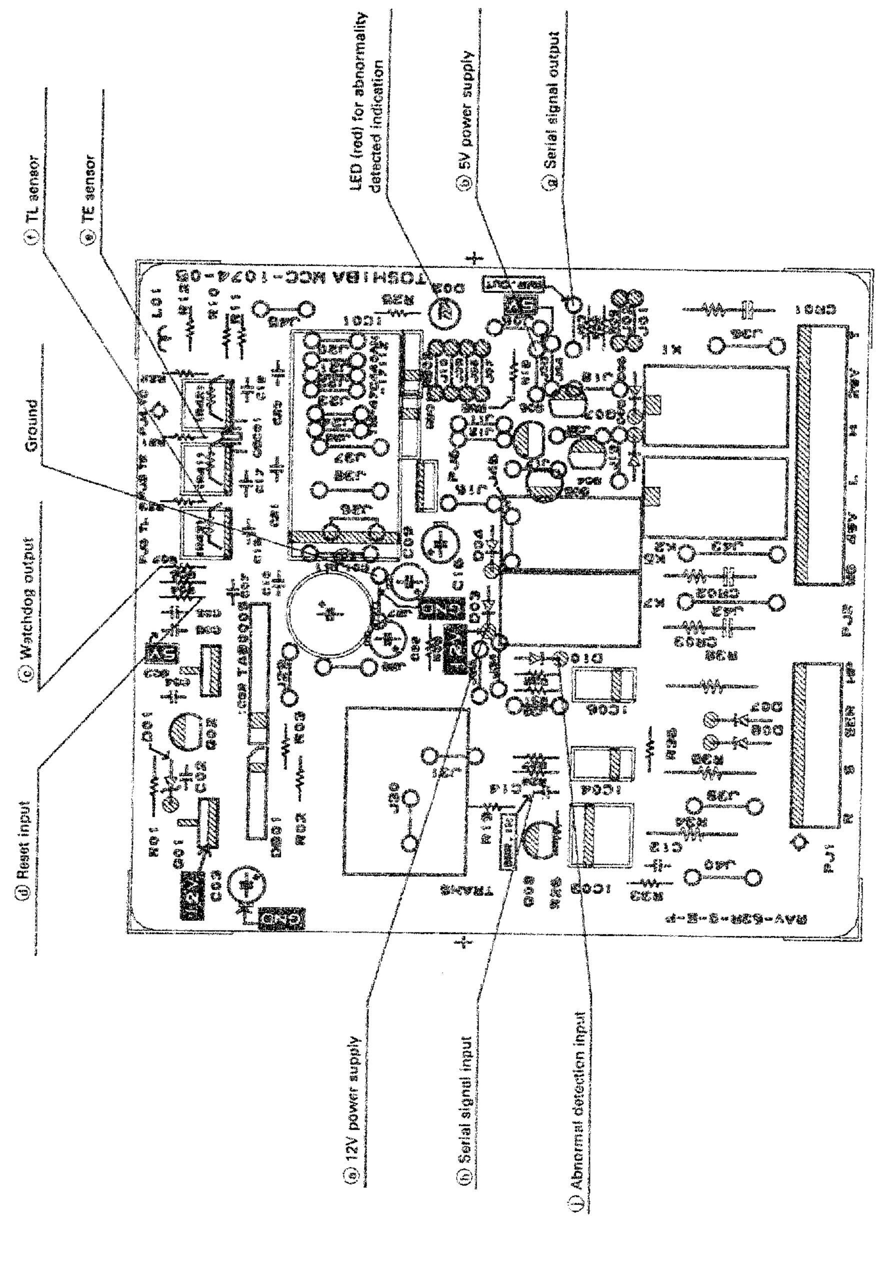
10.1 Outdoor unit control box



10.2 Outdoor unit control circuit diagram

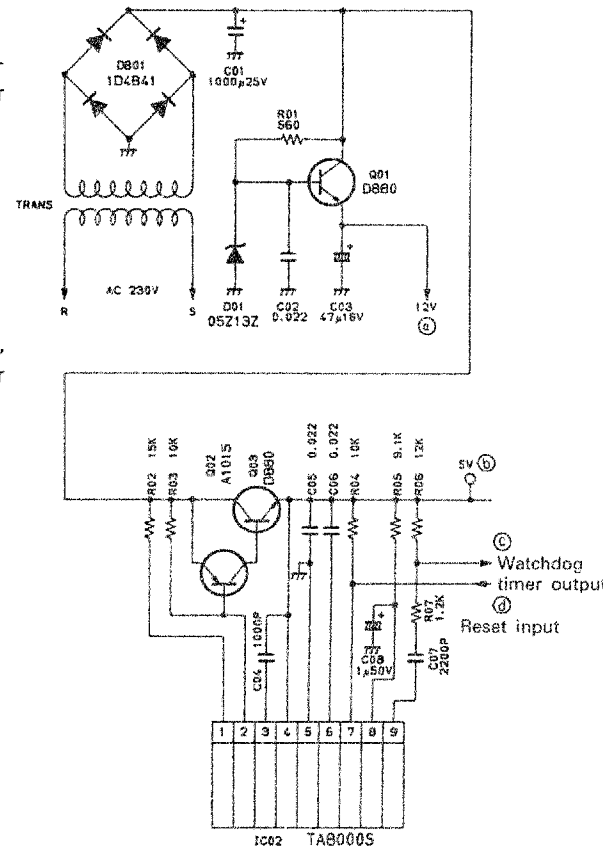


10.3 Outdoor unit PC board



12V power source circuit

Outdoor PC board has a built-in transformer and full-wave rectification by diode bridge (DB01) followed by the provision of transistor (Q01) produces DC power source (ⓐ) at 12V.



5V watchdog timer circuit

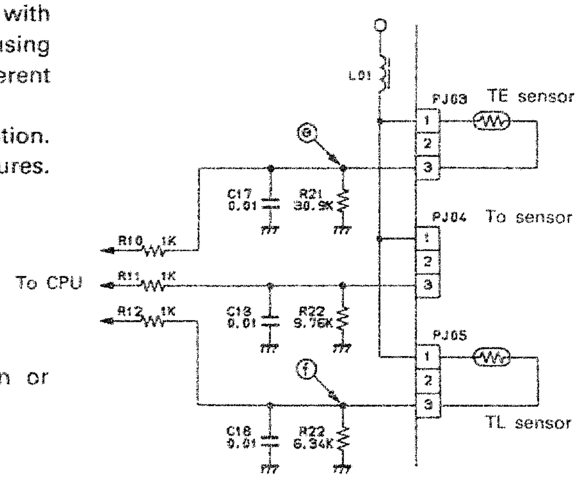
Basically, the same description as the indoor PC board applies, provided, however, that the reset circuit is not added to the outdoor side.

Sensor circuit

This circuit detects the temperature by dividing voltage with resistance and sensor and bringing the voltage value into CPU, using the characteristics of the sensor that resistance varies with different temperatures.
TE is for defrosting, while TL is for low ambient cooling operation.
The following voltages are produced at each of the temperatures.

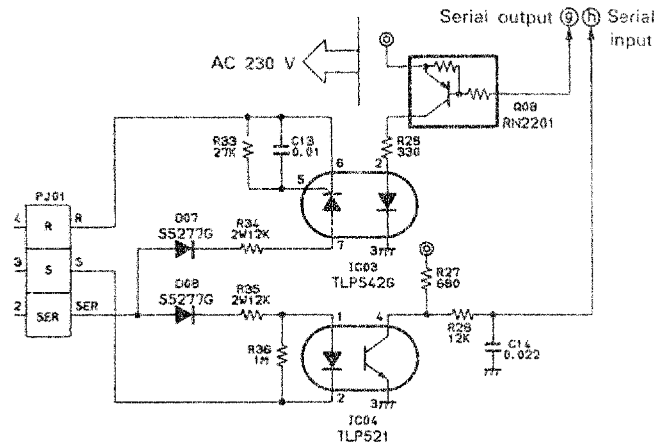
	0°C	25°C
TE ⓔ	2.3V	3.8V
TL ⓕ	0.8V	2.0V

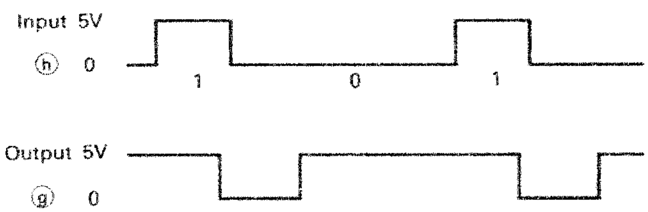
When ⓔ / ⓕ are at GND or 5V, the sensors are in open or short-circuited.



Serial signal circuit (between indoor and outd꜀oor units)

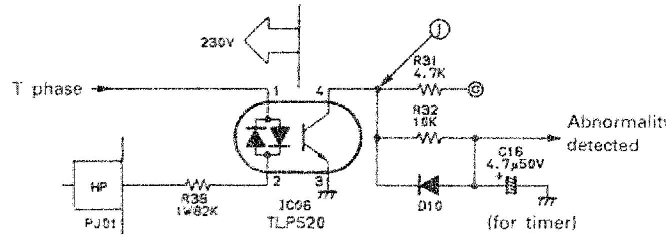
Transmits and receives the signals between indoor and outdoor units in serial signals. As 230V is used for transmitting the signal, the microcomputer section is insulated with photo-coupler with 5V being supplied.

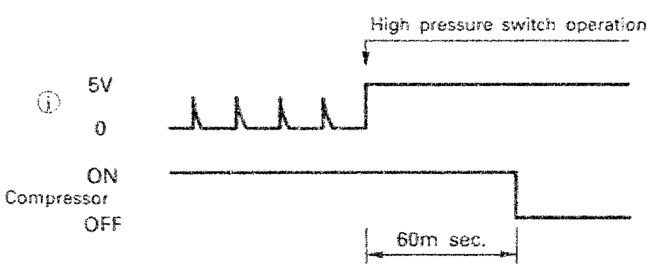




Abnormality-detecting circuit

When high pressure switch is operated, abnormality is detected to stop the compressor.

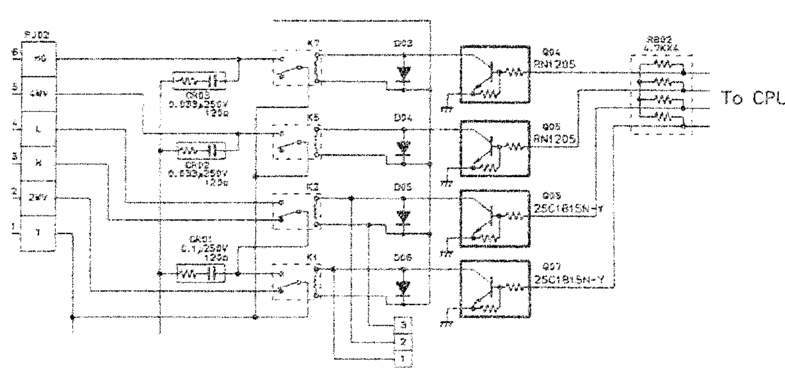




Relay circuit

The relay circuit consists of the diagram in the righthand side.

K1 : Turning fan on and off (2-way valve)
K2 : Changing over H/L of fan
K5 : Turning 4-way valve on and off
K7 : Turning compressor on and off



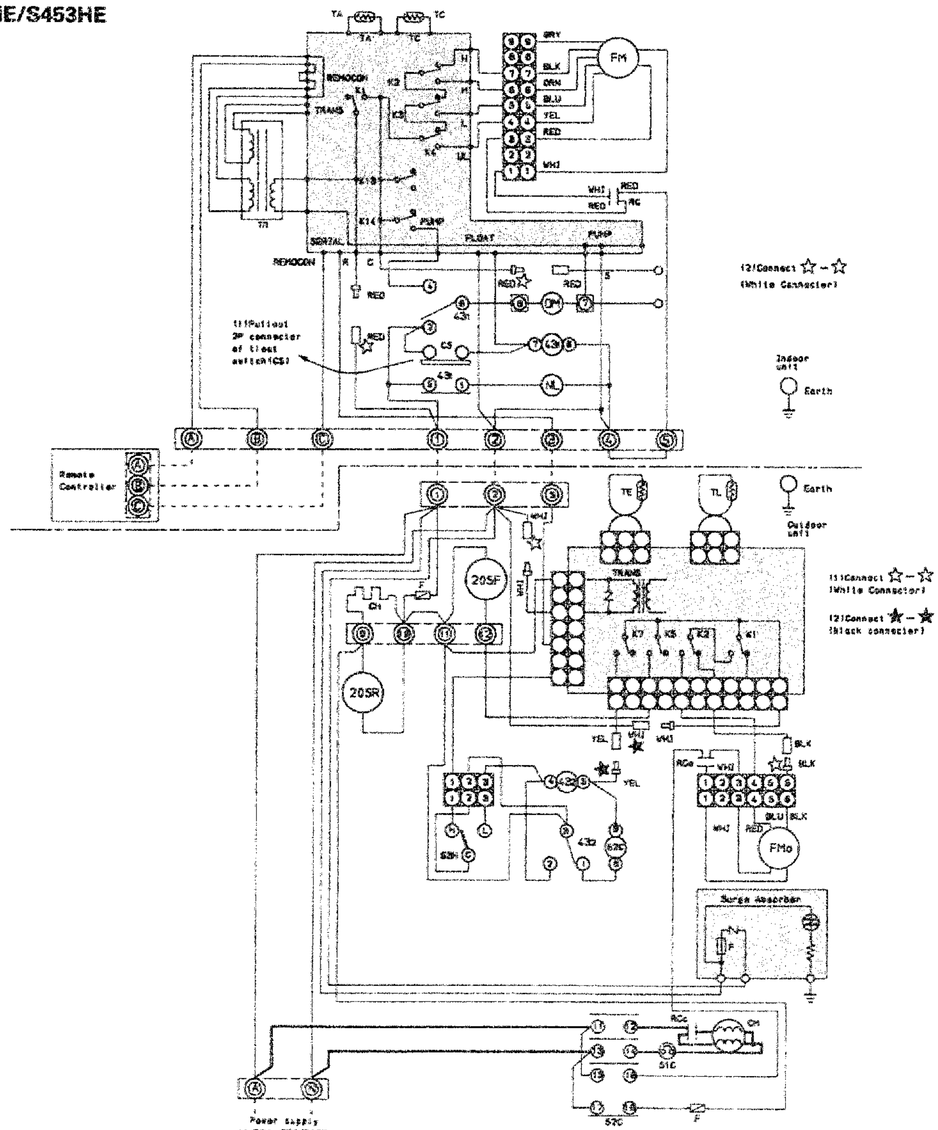
11. EMERGENCY OPERATION (COOLING OPERATION ONLY)

By way of temporary expedient, change-over connectors are incorporated which allow for application of 230V directly to indoor fan motor, outdoor fan motor and magnet switch. Drain pump is operated forcibly by pulling out float switch. In this case, operation and stop is effected by ON/OFF of the power line. (The emergency operation is not provided for heating as it can be substituted by other heating appliances and also because of nonavailability of defrosting approach).

Countermeasures	indoor connector	Pull out the connector of R phase (red) lead wire from pin (1) and connect it with the connector of lead wire for fan motor K1 output (red). Pull out the 2P connector of float switch.
	Outdoor	Pull out 1P white connector for fan motor and connect it with 1P white connector of pin (2). Take out 1P connectors of yellow lead wire and black lead wire for relay output connectors of control PC board and connect yellow lead from relay 43_2 with black lead coming from (6) or white lead coming from (2).
Operation	Operation and stop by the power switch at hand. (High pressure switch becomes the only protective circuit.)	

For the method, refer to next page.

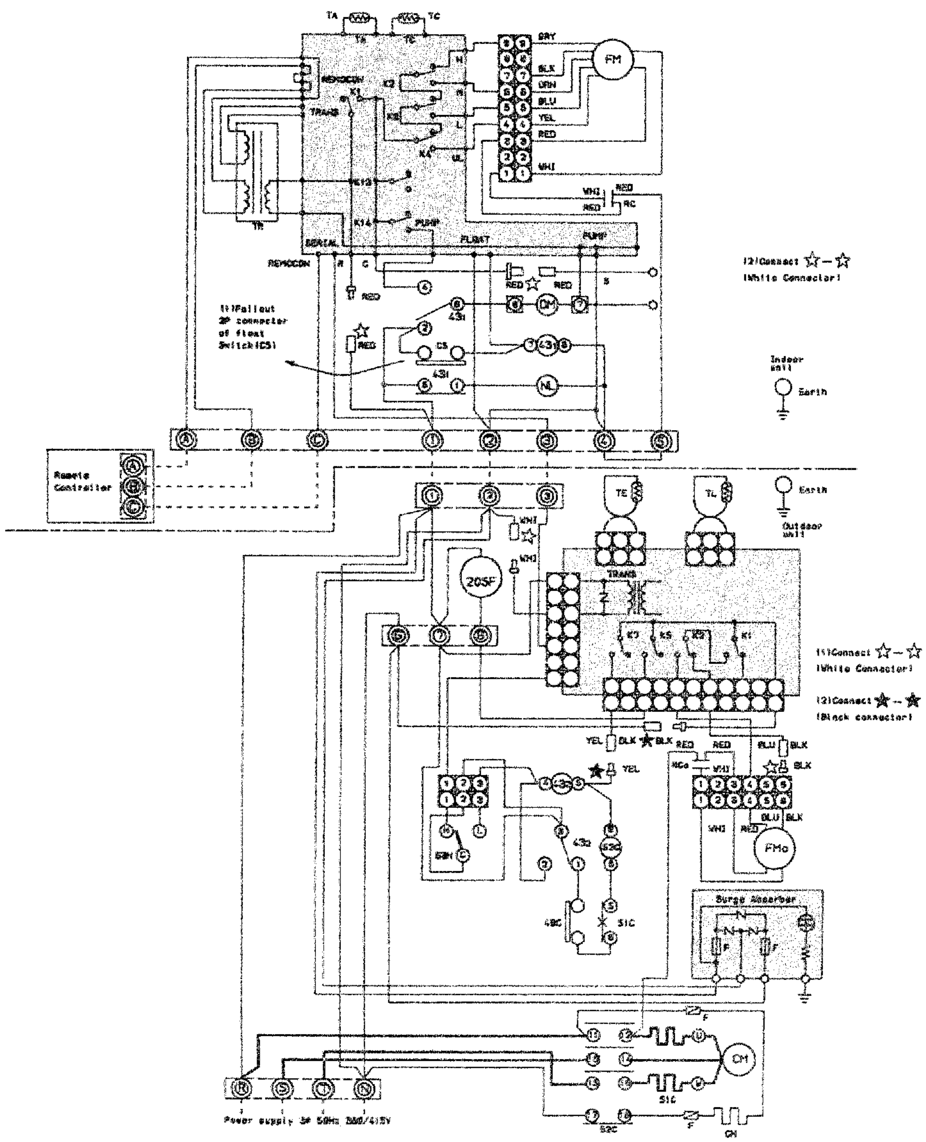
RAV-453UHE/S453HE



◎ shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	**Name**	**Symbol**	**Name**
20SF	Solenoid Coil (4way valve)	CM	Compressor
$K_1 \sim K_{14}$	Relay	52C	Magnetic Contactor
20SR	Selenoid Coil (2way valve)	FM_O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
$43_{1,2}$	Relay	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC_O	Running Capacitor (Outdoor Fan Motor)
FM	Fan motor (Indoor)		
TA	Sensor	RC	Running Capacitor (Indoor Fan Motor)
TC	Sensor		
TR	Transformer	RC_C	Running Capacitor (Compressor)
CS	Float Switch		
NL	Neon Lamp	DM	Drain Motor

RAV-713UHE8/S713HE8



◎ shows terminal block and figures show terminal numbers.
Broken lines show wiring at site.

Don't operate the units with the magnetic contactor pushed.			
Symbol	**Name**	**Symbol**	**Name**
20SF	Solenoid Coil	CM	Compressor
$K_1 \sim K_{14}$	Relay	52C	Magnetic Contactor
49C	Inner Overload Relay	FM_O	Fan Motor (Outdoor)
51C	Overload Relay	TL	Sensor
$43_{1,2}$	Relay	TE	Sensor
63H	High Pressure Switch	F	Fuse
CH	Crank Case Heater	RC_O	Running Capacitor (Outdoor Fan Motor)
TA	Sensor		
TC	Sensor	RC	Running Capacitor (Indoor Fan Motor)
FM	Fan Motor (Indoor)		
TR	Transformer	NL	Neon Lamp
CS	Float Switch	DM	Drain Motor

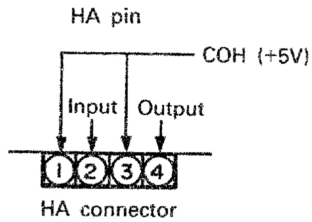
12. APPLIED CIRCUIT

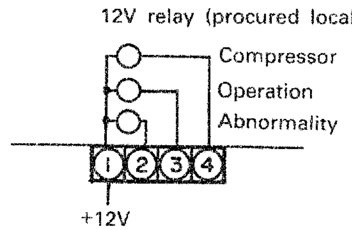
(1) HA terminal (PJ06)

Connectors that can be used for remote start/stop/display are provided on the indoor PC board.

(2) Display output (PJ13)

12V output is provided on the indoor PC board for operation display (interlocked with LED), compressor operation display and outdoor abnormality display.





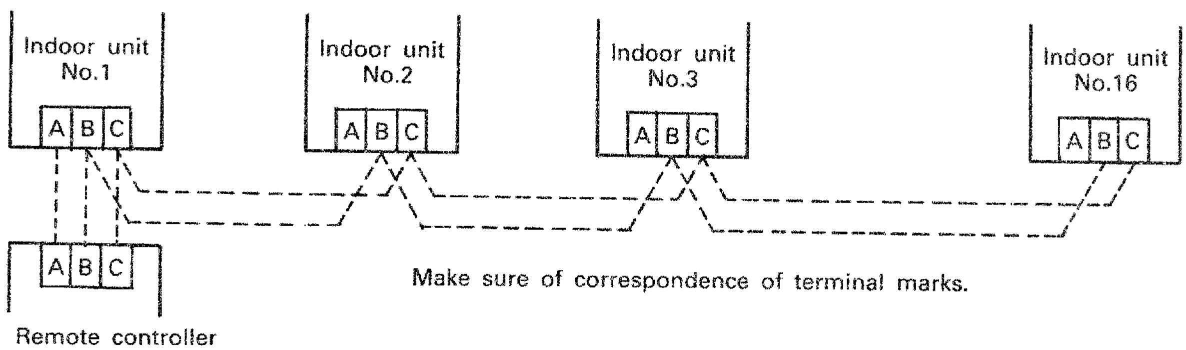
13. WIRING FOR GROUP OPERATION

Up to 16 units can controlled as a group by one remote controller.

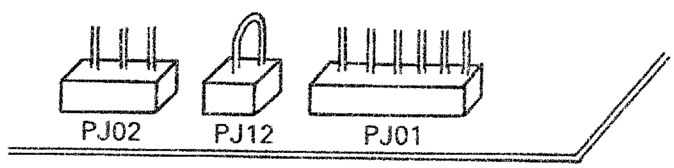
① Align the phase sequence of the power supply to all outdoor units.

② Connect the terminals A, B, C on both of the remote controller and the indoor unit of No1 unit.

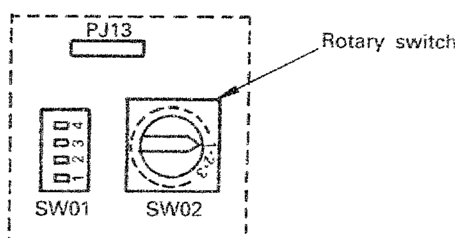
③ Connect terminals B, C on both indoor units of No.1 and No.2 unit. Then connect in the same way No.2 and No.3, No.3 and No.4....up to No.16 unit.



④ Remove the PJ12-counter on the indoor PC boad of No.2 unit and up to No.16 unit to prevent malfunction.

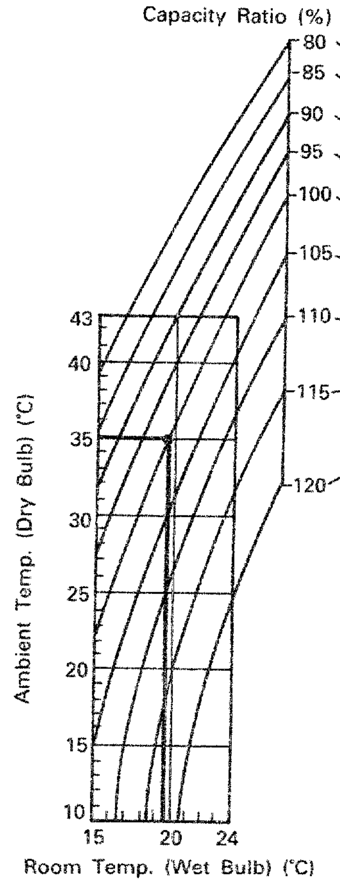


⑥ Set each unit No. rotary switch on the indoor PC board. The unit connected to the remote controller should be set as No.1 unit Then set No.2 and up to No.16 so that start time of each unit is respectively delayed to prevent simultaneous starting current.



14. PERFORMANCE CHARACTER

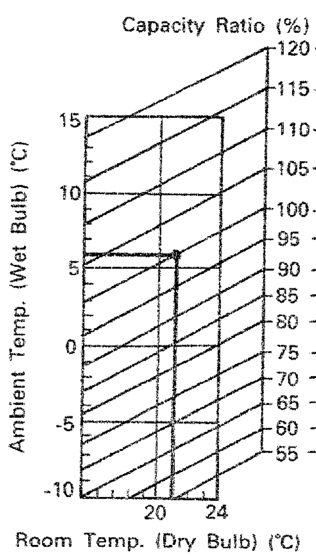
14.1 Cooling capacity



(BTU/h)

RAV-453UHE	RAV-713UHE8
12,800	20,160
13,600	21,420
14,400	22,680
15,200	23,940
16,000	25,200
16,800	26,460
17,600	27,720
18,400	28,980
19,200	30,240

14.2 Heating capacity

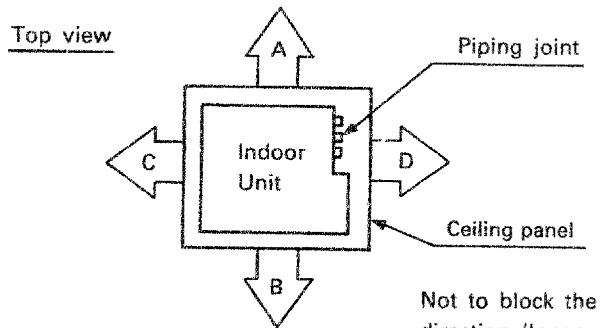


(BTU/h)

RAV-453UHE	RAV-713UHE8
20,640	32,640
19,780	31,280
18,920	29,920
18,060	28,560
17,200	27,200
16,340	25,840
15,480	24,480
14,620	23,120
13,760	21,760
12,900	20,400
12,040	19,040
11,180	17,680
10,320	16,320
9,460	14,960

14.3 Air flow distribution

RAV-453UHE/S453HE
RAV-713UHE8/S713HE8



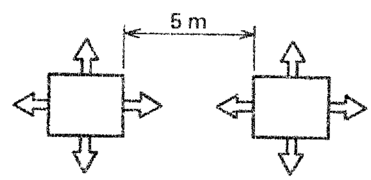
Not to block the A and B direction (longer side).

Air flow direction & volume ratio

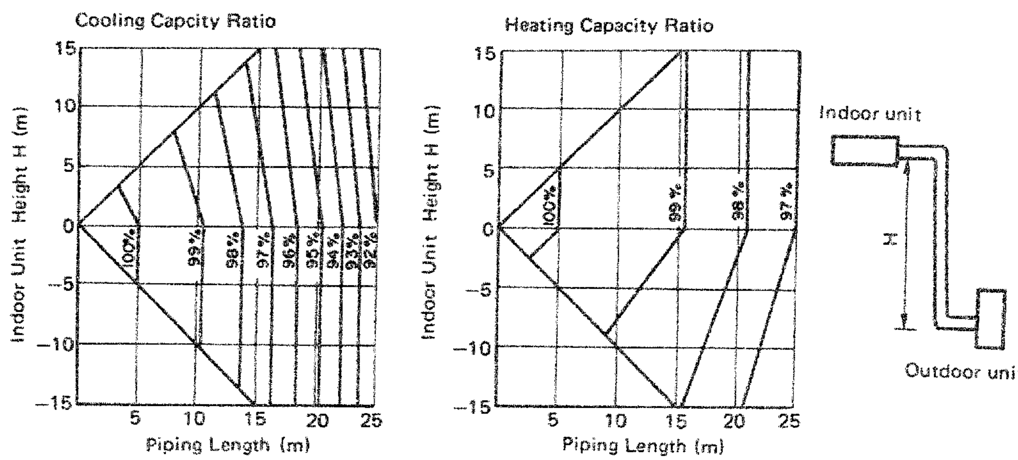
No. of discharge way	4-Way				3-Way			2-Way	
Air flow direction	A	B	C	D	A	B	C	A	B
Air volume ratio (%)	30	30	30	10	33	33	33	50	50

(Note)

Keep 5 m distance between each unit when installing plural units in same room.



14.4 Piping length/cooling capacity/heating capacity



14.5 Piping length/additional refrigerant volume

In case of piping length is more than 5m, less than 25m.

Model	Add. volume (per 1 meter)	Maximum Add. volume
RAV-453UHE	25 g/m	500 g
RAV-713UHE8	35 g/m	700 g

NOTES FOR FLARE PIPING CONNECTION

The performance of the split air conditioner depends greatly on the piping work.
Keep following 5 major points for the piping work.

(1) Do not let dirt and moisture get inside of the pipes
(2) Perform flaring and connection correctly
(3) Purge air from the pipes quickly and efficiently
(4) Check for gas leaks around flare connections
(5) Add refrigerant gas if the piping length exceeds 5m

ADDITIONAL SUPPLY OF REFRIGERANT

On Delivery, the outdoor unit contains the refrigerant for one air purging and 5m piping.
The refrigernt must be supplemented, if the piping length exceeds 5m.
However the piping cannot extended beyond 25m.
The amount of refrigerant to be supplemented varies depending on the type of unit.
Refer to the corresponding instruction manual of each unit to determine the amount of refrigerant for each additional meter.

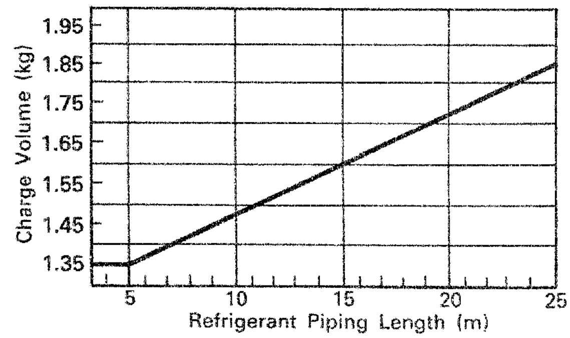
How to calculate the amount of refrigerant to be supplemented (exsample).

M = (L−5) x A (g)

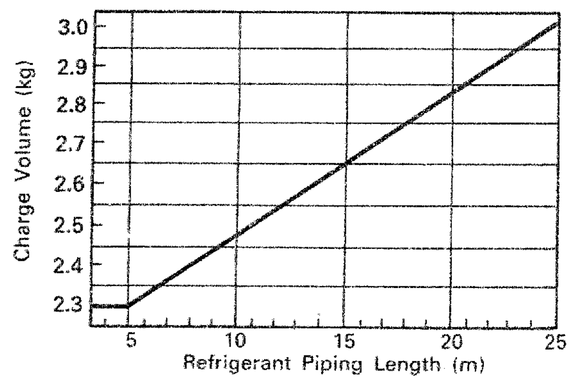
M :	The amount of refrigerant to be supplemented	(g)
L :	The one way actual piping length	(m)
5 :	The standard piping length	(m)
A :	The additional volume of refrigerant for every 1m	(g/m)

Accuracy of additional supply is ±50g.
As an over charge or shortage must give cause for failure of the unit, always measure the precise amount before supply.

RAV-S453HE



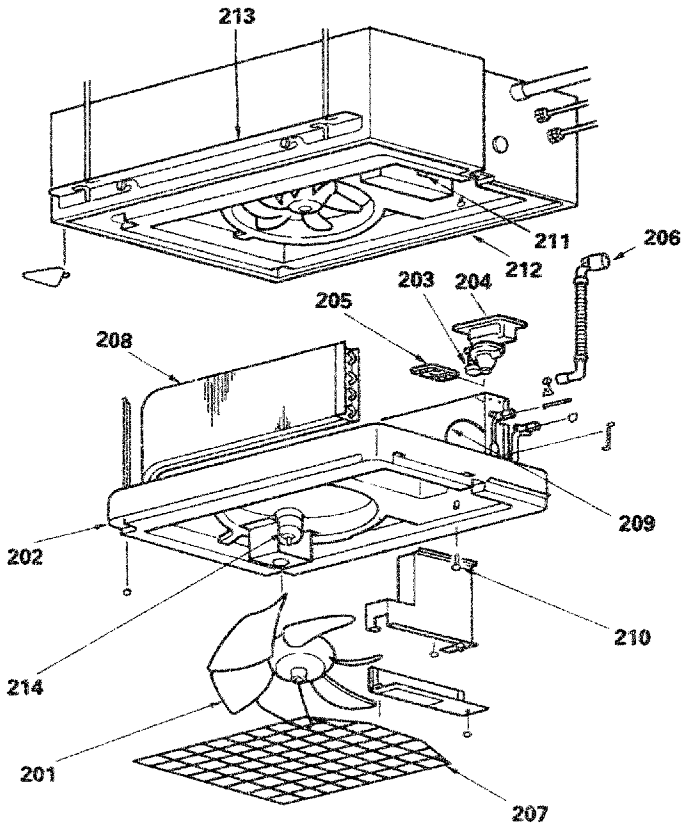
RAV-S713HE8

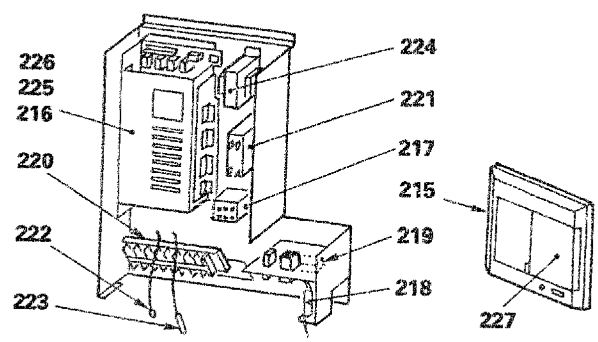


15. EXPLODED VIEWS AND PARTS LIST

15.1 Indoor unit

RAV-453UHE
RAV-713UHE8



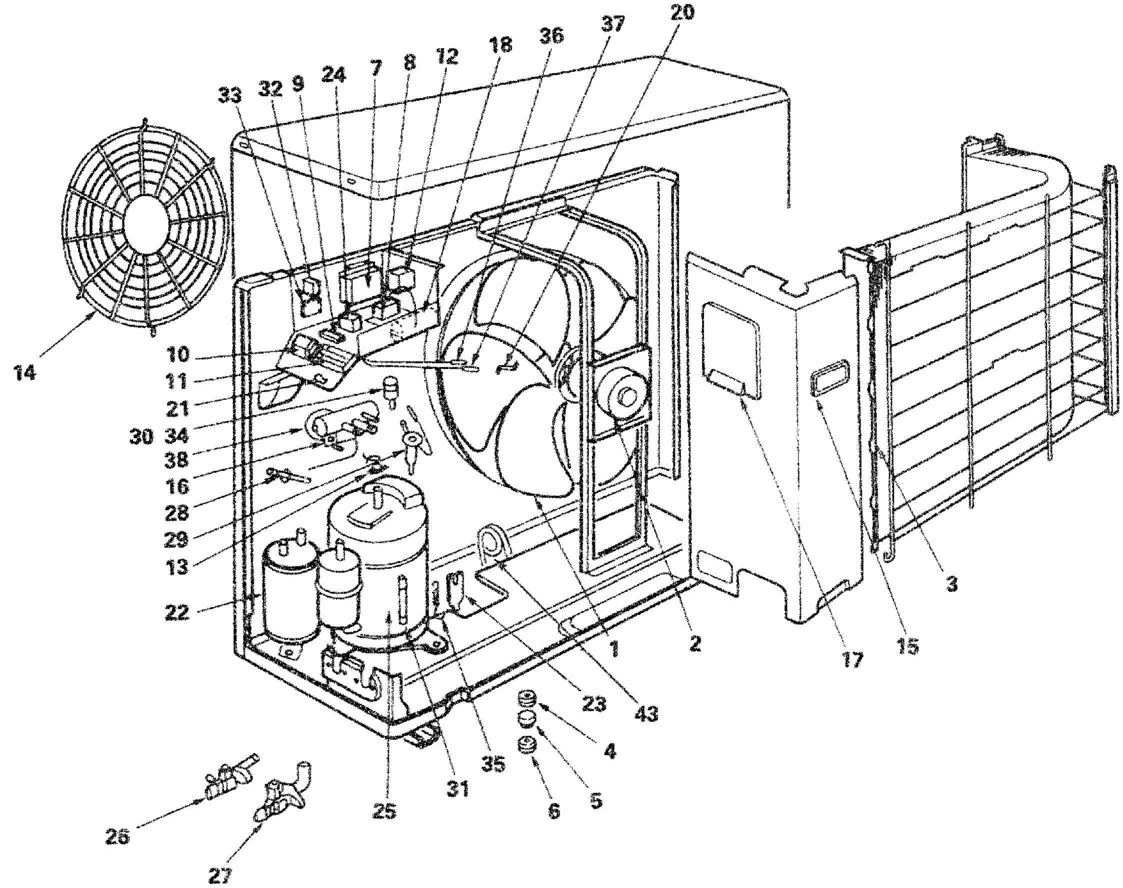


Location No.	Part No.	Description
201	43120153	Fan, Paddle
201	43120163	Paddle Fan
202	43172103	Pan, Drain
203	43151192	Switch, Float
204	43121428	Pump, Drain
205	43179104	Filter, Drain
206	43170186	Hose, Drain
207	43191243	Guard, Fan
208	43144486	Evaporator (RAV-713UHE8)
208	43144509	Evaporator (RAV-453UHE)
209	43047527	Capillary Tube, ID 2.0
210	43179105	Plug
211	43160117	Holder, Power Cord
212	43107206	Hunger, A
213	43107208	Hunger, B
214	43121421	Fan Motor AC 220V 50Hz (RAV-713UHE8)
214	43121423	Motor, Fan, AC 200V 50/60Hz (RAV-453UHE)
215	43169461	Remote Controller

Location No.	Part No.	Description
216	43169461	PC Board (RAV-713UHE8)
216	43169463	PC Board (RAV-453UHE)
217	43154117	Relay
218	43158091	Lamp
219	43160251	Terminal Block, 2P (RAV-713UHE8)
219	43160374	Terminal Block, 2P (RAV-453UHE)
220	43160375	Terminal Block, 8P
221	43155047	Capacitor, Electrolytic (RAV-453UHE)
221	43155084	Capacitor, Electrolytic (RAV-713UHE8)
222	43150118	Sensor
223	43150111	Sensor
224	43158094	Transformer,Power
225	43133011	Spark-Killer
226	43133012	Spark-Killer
227	43162029	Cover

15.2 Outdoor unit

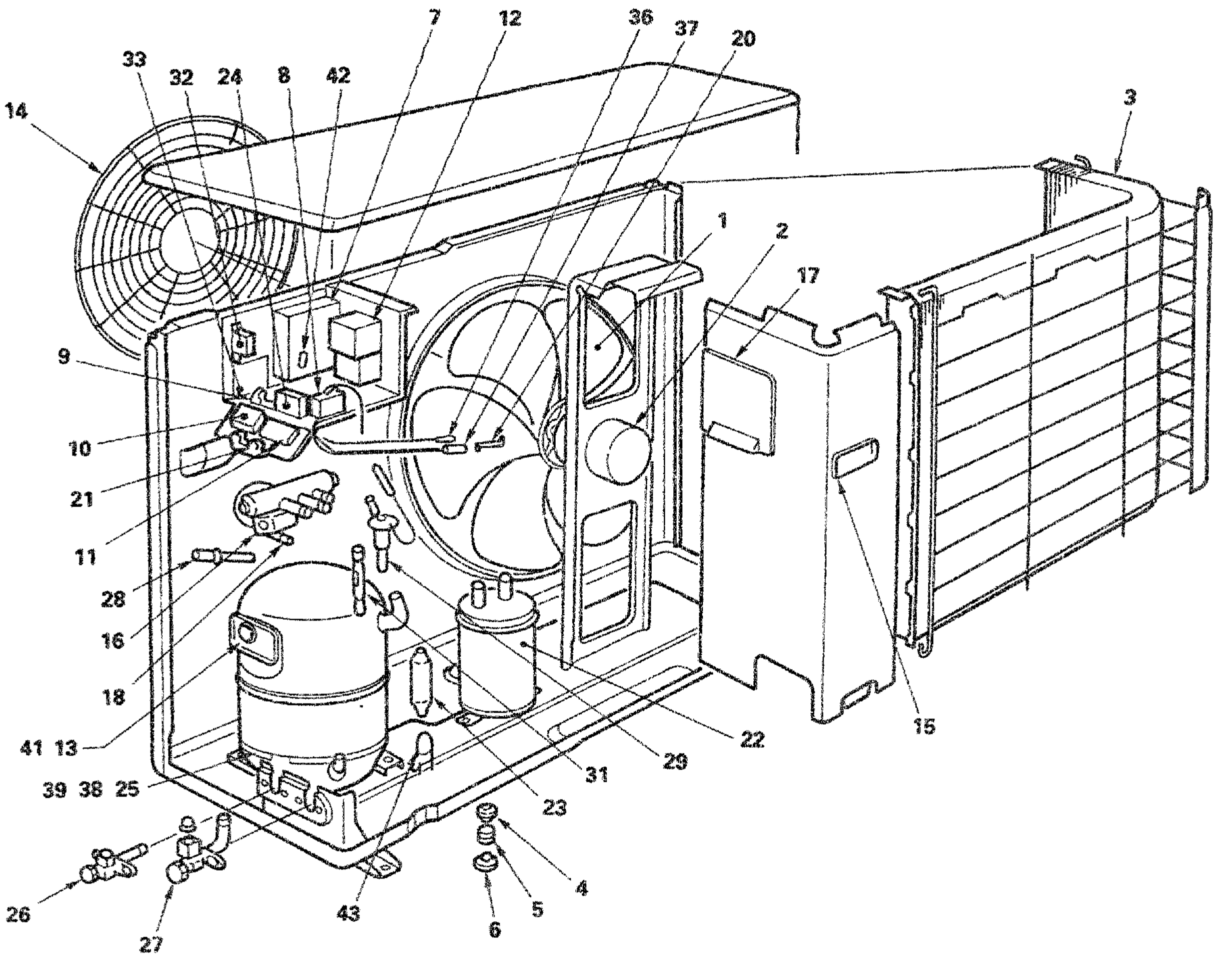
RAV-S453HE



Location No.	Part No.	Description
1	43120156	Fan, Propeller
2	43121473	Motor, Fan, AC 230V 50Hz
3	43143508	Condenser
4	43149212	Base, Spring, A
5	43149198	Spring, Buffer
6	43149208	Base, Spring, B
7	43169464	PC, Board
8	43146387	Switch-High-Pressure
9	43060479	Terminal Block, 4P
10	43160334	Terminal Block, 2P
11	43060750	Terminal Block, 3P
12	43152314	Magnetic Switch
13	43054286	Relay, Over load
14	43191252	Guard-Fan
15	43119368	Hanger
16	43046137	Solenoid Coil, L27-0021
17	43162027	Cover, Electric Parts
18	43055366	Capacitor, Plastic Film, 45MFD 420V
20	43019604	Holder, Sensor
21	43063114	Holder, Cord

Location No.	Part No.	Description
22	43148105	Accumulator
23	43145082	Dryer
24	43155047	Capacitor, Electrolytic, 4MFD, AC 350V
25	43041837	Compressor, AC 220/240V 50Hz PH230X3-4LS
26	43146405	Packed Valve, 1/4 in.
27	43146406	Packed Valve, 1/2 in.
28	43147321	Check Joint
29	43146424	Expansion Valve
30	43046198	Coil, 2 Way Valve
31	43146283	Checked Valve
32	43153003	Relay, SA106H
33	43169465	Surge Absorber
34	43046151	2 Way Valve
35	43193043	Spring
36	43150130	Sensor, Cond. Out
37	43150129	Sensor, Heat Exch.
38	43046009	4 Way Valve, V26-100B
43	43146430	Capillary Tube I.D. 2.4

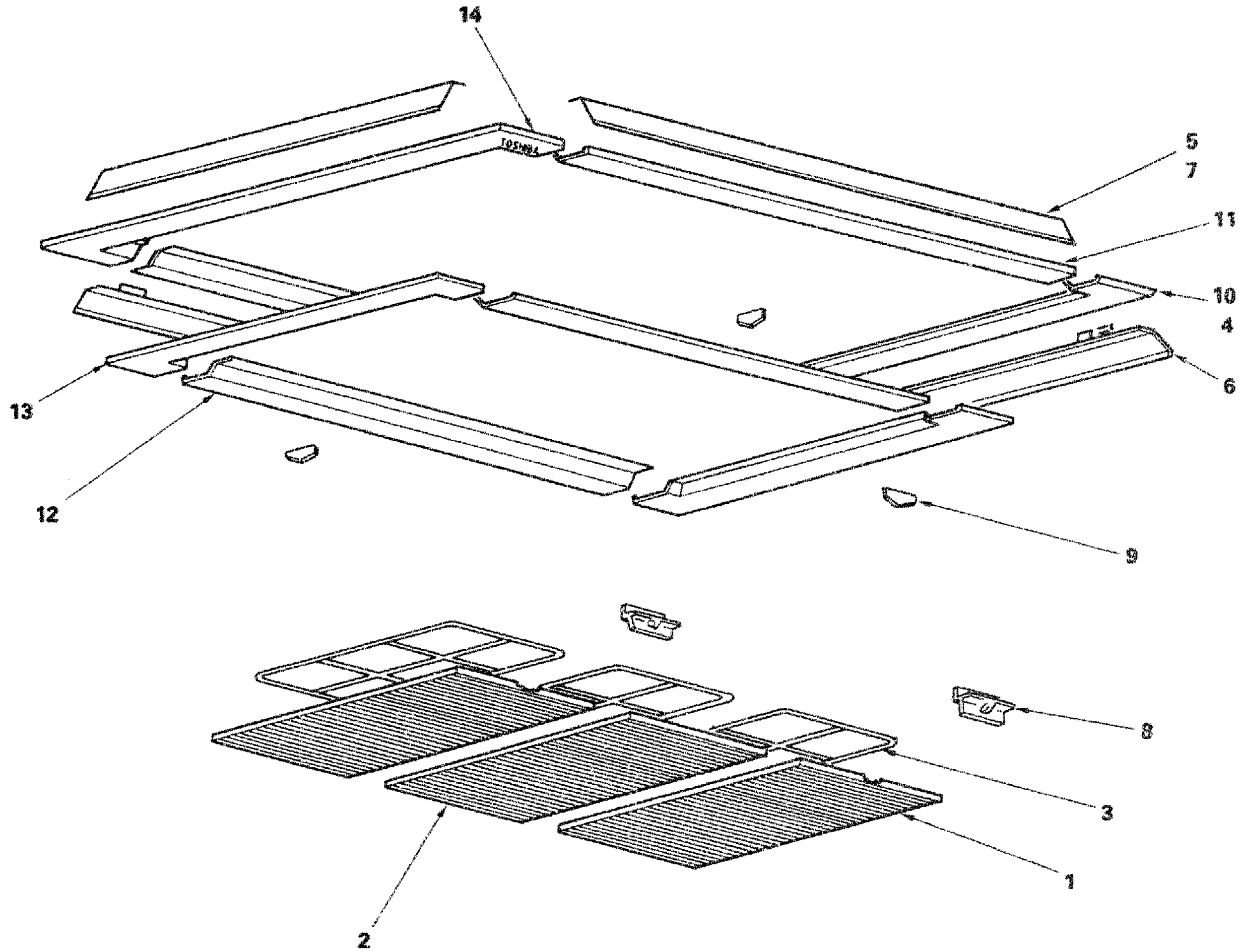
RAV-S713HE8



Location No.	Part No.	Description
1	43120129	Fan, Propeller
2	43121489	Fan, Motor
3	43143505	Condenser
4	43049132	Base, Spring, B
5	43149198	Spring, Buffer
6	43149212	Base, Spring, A
7	43169469	PC Board
8	43146387	Switch-High-Pressure
9	43060157	Terminal Block, 4P
10	43160264	Terminal Block, 2P
11	43060750	Terminal Block, 3P
13	43050244	Thermo, Bimetal
14	43191252	Guard, Fan
15	43119368	Hanger
16	43046137	Solenoid Coil, L27-0021
17	43162027	Cover, Electric Parts
18	43046009	4 Way Valve, V26-100
20	43019604	Holder, Sensor
21	43063114	Holder
22	43148096	Accumulator
23	43145092	Dryer
24	43155080	Capacitor, Plastic Film, 4MFD, 400V
25	43140213	Compressor, YH277JA
26	43146415	Packed-Valve, 3/8 in.
27	43146417	Packed-Valve, 5/8 in.
28	43147321	Check Joint
29	43146291	Expantion Valve
31	43146283	Checked Valve
32	43153003	Relay, SA106H
33	43169438	Surge Absorber
36	43150130	Sensor, Cond, Out
37	43150129	Sensor, Heat, Exch.
38	43193043	Spring
39	43157097	Heater,Crank Case
41	43169254	Holder, Thermostat
42	43060229	Fuse, 1A
43	43146432	Tube Capillary I.D. 2.2

15.3 Celing panel

RBC-U712PAE



Location No.	Part No.	Description
1	43409007	Grille, Inlet
3	43480520	Air Filter
4	43409023	Louver
5	43409024	Louver (RBC-U712PAE)
6	43409008	Louver
7	43409009	Louver (RBC-U712PAE)
8	43401565	Grip

Location No.	Part No.	Description
9	43401564	Stopper, Grille
10	43401568	Panel-A, Outer
11	43401572	Panel-B, Outer (RBC-U712PAE)
12	43409028	Panel-B, Inner (RBC-U712PAE)
13	43409026	Panel-A, Inner
14	43401570	Panel-A, Outer

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